



Coordinated operation of the integrated electricity-water distribution system and water-cooled 5G base stations

Mingyi Sun, Xia Zhao^{*}, Hong Tan, Xinyi Li

State Key Laboratory of Power Transmission Equipment & System Security and New Technology, College of Electrical Engineering, Chongqing University, Chongqing, 400030, China

ARTICLE INFO

Article history:

Received 16 April 2021

Received in revised form

4 September 2021

Accepted 10 September 2021

Available online 14 September 2021

Keywords:

Integrated electricity-water distribution system

Water-cooled 5G base station

Energy & water management

Telecom service provider

Power distribution system

ABSTRACT

To deal with the heavy operational expenditures of the fifth-generation (5G) telecom service providers (TSPs), powering 5G base stations (BSs) with renewable energy (RE) and stimulating a good interaction between the RE-BS and the smart grid is recognized as an effective and practical solution. However, the existing researches on the interaction between RE-BSs and the smart grid mainly study from the perspective of the TSPs, ignoring the operational constraints of the grid. On the other hand, with the rapid development of integrated energy systems, the integrated operation of the power and water distribution system attracts growing attention. And recently, water-cooling has been introduced as a promising energy-saving technique for 5G-BSs, but few efforts have been spent investigating the interaction between the BSs, the power, and the water distribution system. This paper considers the water-cooled 5G-RE-BSs enabled by an integrated electricity-water distribution system (IEWDS), which is managed by the integrated electricity-water service provider (IEWSP). Coordination mechanisms between the TSP and the IEWSP are investigated, and the optimal model with two coordinated operation modes of the IEWDS, incorporating operation constraints of both the IEWDS and the 5G-BS, is proposed. Case studies on two IEWDS sample systems validated the benefits of coordination and the effectiveness of the proposed coordination methods.

© 2021 Elsevier Ltd. All rights reserved.

1. Introduction

1.1. Research motivation

Energy and water support the sustainable development of human society. These two resources are highly interconnected and mutually dependent since the production of energy requires large volumes of water while the water withdrawal, treatment, and distribution are usually energy-intensive processes. For example, the statistics show that the cooling water demand of thermal power plants in the United States exceeds 40% of the total social water intake [1], and the power consumption of the entire Chinese social water cycle exceeds 19% of the total power consumption in the country. Globally, approximately 75% of all industrial water withdrawals were used for energy production in 2014 [2].

Over the past decades, the strong interdependencies between

energy and water have received much attention and developed the concept of water-energy nexus [3]. Numerous researches on water-energy nexus have been carried out from a macro perspective to study such as the life-cycle water use for energy supply, the energy use for water infrastructure, and the water/energy management or saving techniques based on the long-term statistics or at the national level [3–6]. The recent years have seen significant efforts worldwide to develop integrated energy systems (IESs) [7]. The potential synergies between electricity and water are sought to achieve a more efficient, flexible, and decarbonized energy system, which therefore suggests planning and operating power systems and water systems in a more integrated perspective.

On the other hand, fifth-generation (5G) communication has become a hot topic in recent years. Compared with the fourth-generation (4G), 5G promises significant performance improvements in increased data rate, reduced end-to-end latency, and improved coverage, connectivity, security, and reliability, etc. [8,9]. Accordingly, 5G is expected to enable many emerging technologies such as the Internet of Things (IoT), Device to Device (D2D) communications, augmented/virtual reality (AR/VR), self-driving cars,

^{*} Corresponding author.

E-mail addresses: 13767974294@163.com (M. Sun), zx@cqu.edu.cn (X. Zhao), th_199711@163.com (H. Tan), li3xy@mail.uc.edu (X. Li).

Nomenclature*Acronyms*

5G	Fifth-generation
BS	Base station
IES	Integrated energy system
IEWDS	Integrated electricity-water distribution system
IEWSP	Integrated electricity-water service provider
PDS	Power distribution system
RE	Renewable energy
TSP	Telecom service provider
WCS	Water-cooling system
WDS	Water distribution system

Sets and Indices

BS	Bus set with 5G-BSs
i, j	Index of PDS buses
ij	Index of PDS lines
m	Index of WDS nodes
mn	Index of WDS pipes
RES	Bus set with RE source
s	Index of breakpoints of pipe flow rate
t	Index of time-period
x	Index of breakpoints for relative pump speed
y	Index of breakpoints for pump power
Δt	Index of time interval

Parameters

$A_{m,Ta}$	Cross-sectional area of water tanks
$A_{m,Ta,BS}$	Cross-sectional area of BS tanks
$A_{mn,U}$ $B_{mn,U}$	Parameters of pumps
$C_{1,Cut}$	Contract price for electricity deviation
$C_{1,G^*(t)}$	Contract price for electricity
$C_{i,Bat,BS}$	Compensation-coefficient of BS batteries
$C_{i,Deg}$	Battery degradation cost
$C_{i,G(t)}$	Time-of-use electricity price
$C_{i,PVCur}$	Cost-coefficient of PV curtailment
$C_{i,WGCur}$	Cost-coefficient of wind curtailment
$C_{m,S(t)}$	Comprehensive water price
$C_{m,Ta,BS}$	Compensation-coefficient of BS tanks
$E_{i,initial}$	Initial battery energy
$E_{i,initial,BS}$	Initial BS battery energy
$H_{m,initial}$	Initial hydraulic head of water tanks
$H_{m,initial,BS}$	Initial hydraulic head of BS tanks
$H_{mn,U0}$	Shutoff head of pumps
M	A very large positive constant
$P_{1,Pre(t)}$	Desired tie-line power of upper-level grids
$P_{i,0,BS}$	Fixed power of BSs
$P_{i,L(t)}$	Active power load of buses
$P_{i,WCS,BS}$	Power of WCSs
$P_{i,WR,BS}$	Basic power of BSs under work-mode
$P_{i,PV(t)}$	Power output of PV power stations
$P_{i,RES,BS(t)}$	RE source output of BSs
$P_{i,Sleep,BS}$	Basic power of BSs under sleep mode
$P_{i,WG(t)}$	Power output of wind farms
$Q_{i,L(t)}$	Reactive power load of buses
$q_{m,L(t)}$	Water load of nodes
$q_{m,WCS,BS(t)}$	Water-loss rate of WCSs
$q_{mn,P,s}$	Breakpoints of flow rate for pipes
$q_{mn,U,x}$	Breakpoints of flow rate for pumps
r_{ij}, x_{ij}	Resistance and reactance of lines
$r_{mn,P}$	Resistance coefficient of pipes

γ_U	Power factor
ζ_{BSS}	Reduction ratio of basic power demand under sleep mode
η_{Ch}, η_{Dis}	Charging/discharging efficiency-coefficient of battery
$\eta_{Ch,BS}$	Charging efficiency-coefficient of BS battery
$\eta_{Dis,BS}$	Discharging efficiency-coefficient of BS battery
$\eta_{mn,U}$	Efficiency-coefficient of pumps
ρ	Water density
$\omega_{mn,U}, y$	Breakpoints of relative pump speed

Variables and Functions

$am_{n,s(t)}$	Binary variables for linearization
$CBat,BS$	BS battery compensation cost
$CCur$	RE source curtailment cost
C_{cut}	Additional cost(contract)
$CDeg$	Battery degradation cost
CG, CG^*	Cost for power purchase
CS	Cost for water purchase
CTa,BS	BS tank compensation cost
$E_{i,Bat(t)}$	Remaining energy of batteries
$E_{i,Bat,BS(t)}$	Remaining energy of BS batteries
$H_m(t)$	Node head
$H_{m,Ta(t)}$	Hydraulic head of tanks
$H_{m,Ta,BS(t)}$	Hydraulic head of BS tanks
$lij(t), ui(t)$	Auxiliary variables of SOCP relaxation
$P_{i,BS(t)}$	Total power demand of BSs
$P_{i,Ch(t)}$	Charging/discharging power of batteries
$P_{i,Dis(t)}$	in the PDS
$P_{i,Ch,BS(t)}, P_{i,Dis,BS(t)}$	Charging/discharging power of BS batteries
$P_{i,G(t)}$	Active power output of power plants
$P_{i,PVCur(t)}$	PV curtailment
$P_{i,RESCur,BS(t)}$	RE source curtailment of BSs
$P_{i,U(t)}$	Active pump power
$P_{i,WGCur(t)}$	wind curtailment
$P_{ij(t)}$	Active power flow of lines
$Q_{i,G(t)}$	Reactive power output of power plants
$Q_{i,U(t)}$	Reactive pump power
$Q_{ij(t)}$	Reactive power flow of lines
$q_{m,BS(t)}$	Total water demand of BSs
$q_{m,In(t)}$	Inflow rate of water tanks
$q_{m,Out(t)}$	Outflow rate of water tanks
$q_{m,In,BS(t)}$	Inflow rate of BS tanks
$q_{m,Out,BS(t)}$	Outflow rate of BS tanks
$q_{m,S(t)}$	Flow rate of water sources
$q_{mn,P(t)}$	Flow rate of pipes
$q_{mn,U(t)}$	Flow rate of pumps
$vi(t)$	Voltage magnitude of buses
$\Delta H_{mn,P(t)}$	Hydraulic head difference of pipes
$\Delta H_{mn,U(t)}$	Pump head gain
$\Delta \hat{H}_{mn,P(t)}$	Hydraulic head difference of pipes in piecewise linearization
$\Delta \hat{H}_{mn,U(t)}$	Pump head gain in piecewise linearization
$\hat{P}_{mn,U(t)}$	Pump power in piecewise linearization
$\lambda_{mn,s(t)}$	linearization
$\lambda_{mn,x,y(t)}$	Continuous variables for piecewise linearization
$\mu_{i,Bat(t)}$	0-1 variables of batteries
$\mu_{i,Bat,BS(t)}$	0-1 variables of BS batteries
$\mu_{m,Ta(t)}$	0-1 variables of water tanks
$\mu_{m,Ta,BS(t)}$	0-1 variables of BS tanks
$\omega_{mn,U(t)}$	Relative pump speed

and Internet of Vehicles (IoVs) [8,10]. However, these performance improvements bring about the obvious issues of significantly increased energy demand and operational expenditures, which challenges 5G telecom service providers (TSPs) greatly and may become the major deployment obstacle of 5G systems. Evidence suggests that base stations (BSs) are the most energy-intensive components of a wireless network. Nearly 80% of the energy required by the networks is consumed at BSs [11]. The 5G communication uses a high-frequency millimeter-wave (mmWave) to carry data. Due to the strong absorption effect of the atmosphere on mmWaves and the fact that mmWaves are easily blocked by walls, foliage and even peoples' bodies, the coverage of a 5G-BS is much smaller than that of the previous 2G/3G/4G [10,12]. The typical density of the BS for 4G is around 8–10 BS/km², while that of 5G is anticipated to be 40–50 BS/km² [12]. Besides, 5G-BSs are generally equipped with some new high-energy-demand components such as mmWave transceivers, fast data converters, integrated multiple-input-multiple-output (MIMO) antennas, and high-power amplifiers [13]. It is estimated the energy consumption of each 5G-BS grows up to twice or more than that of a 4G-BS [13]. Therefore, the total energy consumption of 5G-BSs might be 9 times or more than that of 4G to provide the same service coverage [10]. A survey estimated that the high-density BSs and the additional equipment required by 5G will increase the TSPs' total network energy consumption by up to 150%–170% by 2026 [14].

To address the energy consumption issue, a variety of energy-saving solutions are suggested to improve the energy efficiency (EE) of 5G networks such as resource allocation, relaying techniques, multiple radio access, cell breathing, adaptive MIMO switching, BS sleeping, and so on [15,16]. Besides, powering BSs with renewable energy sources such as photovoltaic (PV) panels and wind turbines is considered a promising and practical energy-harvesting solution to reduce the operational expenditure of TSPs [13]. Statistics in Ref. [17] show that renewable-energy-powered BSs (RE-BSs) have grown from 9000 in 2010–43,000 by mid-2014. To deal with the intermittence and fluctuations of renewable energies, RE-BSs are generally equipped with batteries or connected to a power grid for reliable energy supply. In such cases, the RE-BSs are powered by hybrid sources, which gives rise to the problem of the power coordination or energy management of BSs. Specifically, as conventional power grids evolve into the smart grids with the main features of distributed generation, demand-side management, and price signaling [13,18], more power coordination problems appear since the RE-BS connected to a smart grid may act as a prosumer to participate in the energy trading and demand-side response programs with the smart grid.

In addition to the above-mentioned energy-saving solutions, water-cooling has been introduced to reduce the cooling costs of BSs [19]. Compared with traditional air cooling, water-cooling has the advantages of higher heat transfer coefficients, low acoustic noise, environmental friendliness, and a high possibility of heat reuse [20]. Although water-cooling for BSs has not been widely applied in commercial applications, it is gaining more and more attraction. For example, in 2018, Nokia has cooperated with Finnish telecom operator *Elisa* and power system provider *Efore* to deploy the world's first water-cooled 5G-BS in Helsinki, Finland [21]. Moreover, there are many successful applications of water-cooling for data centers [22,23], which provide excellent references for the design and operation of water-cooled BSs. In this work, cooling water tanks are installed for BSs and supplied by the WDS to achieve circulative cooling and water demand regulation.

With the RE sources, batteries, and water tank installed, the 5G-BS is essentially a small-scale integrated water-energy system. It acts as a distributed energy-prosumer to the power grid and a flexible water load to the WDS. Accordingly, the interdependencies

among the 5G-BS, the power grid, and the WDS are necessary to be recognized, and this is the motivation behind the current work.

1.2. Literature review

1.2.1. Coordinated operation of power systems and water systems

Researches on the coordinated operation of power systems and water systems are briefly reviewed as follows.

An optimal power-water flow model of an integrated electricity-water distribution system (IEWDS) considering network constraints of both power distribution systems (PDSs) and water distribution systems (WDSs), and the pump coupling is proposed in Ref. [24]. In Ref. [25], an optimal model for operating an IEWDS is proposed to minimize the operation cost of the integrated electricity-water service provider (IEWSP) and enhance the wind power accommodation. An analytical framework based on time-series optimal power and water flow simulation is developed in Ref. [26] to analyze the interdependencies between power systems and WDSs. In Ref. [27], the flexibility offered by WDSs to the power systems is studied, and a day-ahead power system dispatch model integrating the energy regulating capacity of WDSs is proposed. Based on this, the authors of [27] further propose a coordinated operation model for the IEWDS incorporating the flexibility provided by water treatment and desalination plants in Ref. [28]. A dispatch method for a combined electricity-water supply system is proposed and applied to smart communities in Ref. [29], whereas the dispatch model with WDS constraints proposed in Ref. [30] studies the optimal operation of islanded water-energy microgrids. In addition to the mentioned work [24–30] focusing on the linkage between electricity and water, the interdependencies between the water, thermal and electrical energy are studied with an island electricity-water-heat microgrid model proposed in Ref. [31], the IES model composed of electricity, heating, and water distribution systems is presented to improve the wind power accommodation in Ref. [32], and the coordinated operation for an integrated water and energy system including electricity, gas, and water distribution systems is investigated in Ref. [33].

Although existing studies have made some contributions towards the coordinated operation of power systems and water systems, 5G-BSs and their operation flexibilities are not incorporated in coordinated operation modeling, and therefore interdependencies among the 5G-BS, the power grid, and the WDS are not addressed.

1.2.2. Interaction between RE-BSs and smart grids

Existing studies on the interaction between RE-BSs and smart grids are briefly reviewed in this section.

Smart grid enabled mobile networks are considered in Ref. [34] where energy from the grid and RE sources in each BS are assumed to be shared among the BSs, and an optimization problem for joint BS operation and power distribution is formulated to minimize the on-grid power consumption. In Ref. [35], RE-BSs are equipped with batteries and allowed to sell energy to a smart grid. In Ref. [36], the RE-BS with a dual-battery architecture is studied and the joint scheme for user association and smart grid energy transfer is proposed. In Refs. [37,38], the BS utilizes its harvested RE and stored energy to provide ancillary services to a smart grid by responding to the requests of increasing or decreasing the on-grid power. Accordingly, energy management strategies for BS on/off switching and activating/deactivating of radio resource blocks are proposed in Refs. [37,38], respectively.

However, all the above studies in Refs. [34–38] investigated the interaction between RE-BSs and smart grids mainly from the perspective of communication networks or TSPs. They focus on the optimal operation of BSs, and the smart grids are treated as ideal

power sources, i.e., operation constraints of the grids are ignored. In fact, with the rapid development of 5G, the high density of 5G-BSs and their huge energy consumption challenge the operation of both 5G networks and the grids; while on the other hand, the coordinated operation of 5G-BSs is expected to bring more aggregation benefits to the grids. Therefore, it is necessary to model and analyze the interaction between BSs and the grids in a more integrated way where constraints of both BSs and the grids are considered.

1.3. Contributions

In this work, we consider the *water-cooled 5G-BS with RE source* (abbreviated as 5G-BS below) embedded in an IEWDS. The novelty of this work is that we highlight the interaction between the 5G-BS and the IEWDS from the perspective of energy & water management. To our best knowledge, such interaction has not been investigated in the literature.

In contrast to the previous work in Refs. [34–38], we investigate the interaction between the BS and the IEWDS from the perspective of energy management rather than the communication service. Although there are some recent studies on 5G in the field of power and energy system (PES) such as [12,39], they focus on the 5G enabled PES or exploring the application potential of 5G techniques in PES.

To address the energy & water interaction between the 5G-BS and the IEWDS, the cooperation mechanisms between the TSP and the IEWSP are developed, and the corresponding optimal model, incorporating detailed operation constraints of both the IEWDS and the 5G-BS, are formulated.

The main contributions of this work are as follows.

- (1) *An advanced water-cooled 5G-BS is modeled for energy & water management.* The water-cooled 5G-BS with RE sources, battery, and water storage is modeled as a small-scale integrated electricity-water system from the perspective of energy & water management.
- (2) *The energy & water interaction between the 5G-BS and the IEWDS is studied and the coordination mechanisms are proposed.* Two possible win-win operation modes of the IEWDS are developed to explore the coordination mechanisms between the TSP and the IEWSP. The first mode focuses on the coordination between the IEWSP and the TSP, and the second one is a dual coordination mode, which features the coordination between the IEWSP and the upper-level grid, and the coordination between the IEWSP and the TSP.

- (3) *The optimal model to address the coordination mechanisms between the TSP and the IEWSP is formulated.* The coordination mechanisms are conducted by an optimal model with different objectives and detailed operation constraints of both the 5G-BS and the IEWDS. The SOCP relaxation and piecewise linearization techniques are applied to reformulate the proposed model into a tractable mixed-integer second-order cone programming (MISOCP) problem.

2. IEWDS and 5G-BS

2.1. Interdependent structure

The IEWDS embedded with 5G-BSs includes two service providers, i.e., the IEWSP and the TSP, as shown in Fig. 1. The IEWSP owns the PDS, the WDS, electrical pumps, PV stations, and wind farms, and is responsible for their operation and maintenance. The TSP has the ownership of all the 5G-BSs in the IEWDS as well as the operation and maintenance authority.

The 5G-BS is mainly composed of the communication devices, the RE source, the battery, and the water-cooling system (WCS) along with the controllable water tank [19,22], as shown in Fig. 1(a). The water tank can be installed inside the BS to provide an on-site water supply, or outside the BS to supply water to multiple BSs within an area. Note that the 5G-BS are supposed to be able to feed surplus electricity back to the PDS, and a similar bidirectional water flow between the 5G-BS and the WDS is assumed.

The power demands of the 5G-BS include the transmitting power of the communication devices, the fixed power demand of the BS, and the power supplied to the WCS. The power demands can be supplied by the purchased power from the IEWSP, the RE source, and the BS battery. The transmitting power is the power required to provide services to the telecom users of the BS. Based on the test results of [40], the transmitting power of 5G-BSs is nearly proportional to the communication load. The fixed power is the basic power needed to operate a 5G-BS, which does not vary with the communication load. The power supplied to the WCS is used to circulate cooling water and cool the energy-intensive 5G devices of the BS.

As shown in Fig. 1(b), the IEWSP purchases electricity and water from the upper-level power system and the reservoir/water plant to satisfy the power and water demand throughout the IEWDS, respectively. Electrical pumps are the main coupling facilities between the PDS and the WDS, which consume power to boost the hydraulic pressure of the entire or partial WDS. In Fig. 1(b), the

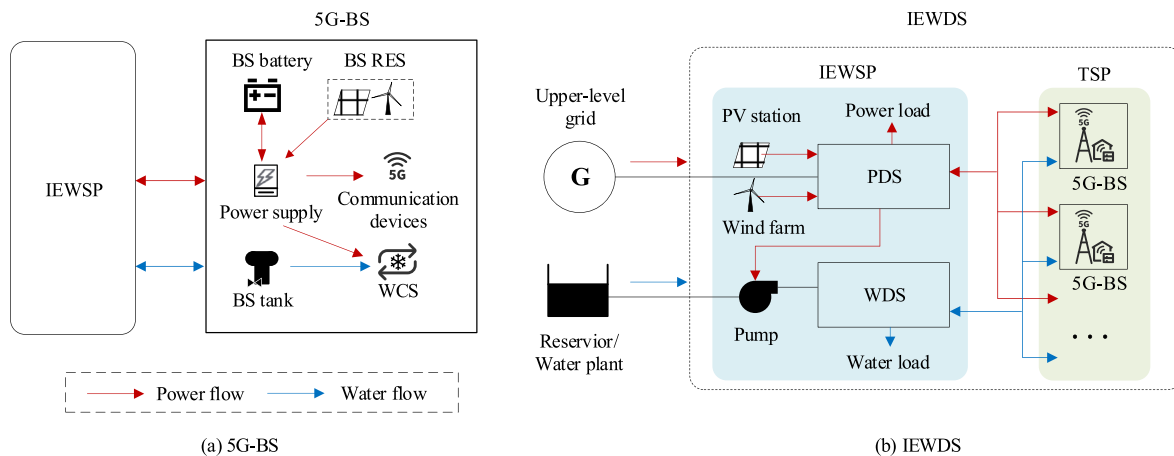


Fig. 1. The schematic diagram of the 5G-BS and the IEWDS.

electrical pump is directly connected to the reservoir/water plant to control the hydraulic pressure, and small booster pumps may also be installed in the WDS to provide secondary pressurization. Besides BS batteries and BS tanks, both the PDS and the WDS are equipped with storages to provide electricity and water backups for the operation of the IEWDS.

2.2. Coordination mechanisms

When the IEWSP and the TSP operate independently, the BS battery is only used for the emergency power supply of the telecom equipment and the BS tank is responsible for replenishing the water loss during the cooling water circulation. Considering that power emergency rarely occurs and water loss is generally negligible, these two kinds of storage equipment are almost idle in the normal operation of the IEWDS.

To make full use of the BS batteries and water tanks and to create synergies between the operations of the IEWDS and 5G-BSs, two coordination mechanisms, as shown in Fig. 2., are developed in this work. The first mechanism (Fig. 2(a)) focuses on the coordination between the IEWSP and the TSP; and the second one (Fig. 2(b)) is a dual coordination mode, which features the coordination between the IEWSP and the upper-level grid, and the coordination between the IEWSP and the TSP.

2.2.1. Mechanism 1

As shown in Fig. 2(a), Mechanism 1 assumes that the control authority of the BS battery and BS tank is transferred from the TSP to the IEWSP, i.e., BS batteries and BS tanks are dispatchable resources to the IEWSP. Under this mechanism, the TSP provides the BS battery and BS tank to the IEWSP in exchange for service compensation, and the IEWSP has the potential to lower its operation cost by utilizing these dispatchable resources. Therefore, a win-win situation may be achieved.

2.2.2. Mechanism 2

Mechanism 2 is a dual coordination mode, as shown in Fig. 2(b). Besides the coordination between the IEWSP and the TSP, which is similar to Mechanism 1, the coordination between the IEWSP and the upper-level grid is expected. The upper-level grid sends the daily tie-line power curve to the IEWDS to inform its desired tie-line power, and the IEWSP tries to operate the PDS, the WDS, and the BS batteries and tanks coordinately to follow this desired curve. By doing this, the IEWSP is compensated by low-priced electricity from the upper-level grid. On the contrary, the IEWSP will be charged a penalty for the deviation in the actual tie-line power curve from the desired one.

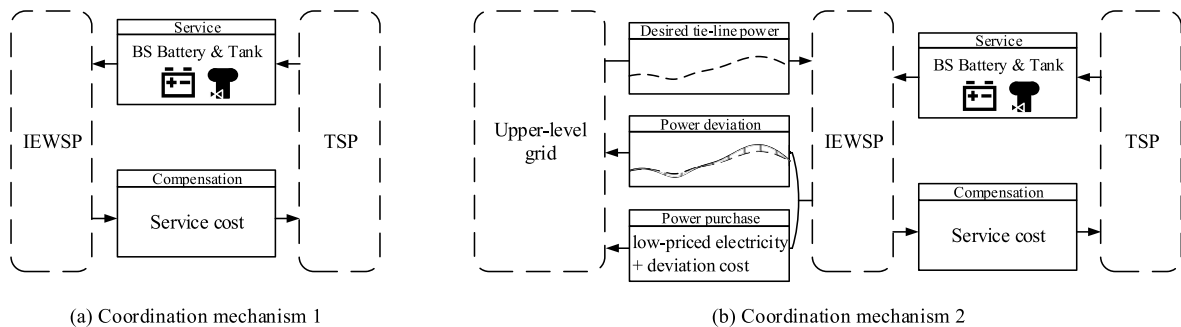


Fig. 2. Two coordination mechanisms.

3. Problem formulation

In this section, the above two coordinated mechanisms will be formulated as two optimal problems. The IEWSP is the decision-maker for both problems. Different objectives are designed for the two mechanisms, while the same constraints, including all the operational constraints of the PDS, the WDS, the coupling facilities between the PDS and the WDS, and 5G-BSs, are satisfied.

3.1. Modeling of 5G-BS

1) Power balance

According to the schematic diagram of the 5G-BS shown in Fig. 1 (a), the power balance of the 5G-BS is

$$P_{i,BS}(t) = P_{i,Re,BS}(t) + P_{i,Ch,BS}(t) - P_{i,Dis,BS}(t) + P_{i,RESCur,BS}(t) - P_{i,RES,BS}(t) \quad (1)$$

The sleep mode of 5G RE-BS [41] is considered in this work. Fig. 3(a) and Fig. 3(b) depict the power demand of the 5G-BS in the work mode and the sleep mode, respectively. The WCS power and the fixed power comprise the basic power of the 5G-BS. In Fig. 3, $P_{i,WR,BS}$ and $P_{i,Sleep,BS}$ represent the basic power of the 5G-BS in the work mode and the sleep mode, respectively [42]. The power equation of the two modes is formulated as (2).

$$P_{i,Re,BS}(t) = \begin{cases} P_{i,WR,BS} + P_{i,L,BS}(t) & \text{if } P_{i,L,BS}(t) \neq 0 \\ P_{i,Sleep,BS} & \text{if } P_{i,L,BS}(t) = 0 \end{cases} \quad (2)$$

$$P_{i,WR,BS} = P_{i,0,BS} + P_{i,WCS,BS} \quad (3)$$

$$P_{i,Sleep,BS} = (1 - \zeta_{BSS})P_{i,0,BS} + P_{i,WCS,BS} \quad (4)$$

$$0 \leq P_{i,L,BS}(t) \leq \bar{P}_{i,L,BS} \quad (5)$$

$$0 \leq P_{i,RESCur,BS}(t) \leq P_{i,RES,BS}(t) \quad (6)$$

Under work mode, the basic power is the sum of the fixed power $P_{i,0,BS}$, and the WCS power $P_{i,WCS,BS}$, as shown in (3), while under sleep mode, the BS reduces the transmitting power to zero and the fixed power demand to some degree (generally 15%–20% [11], which is reflected by the coefficient ζ_{BSS} in (4)). Equations (5) and (6) restrict the transmitting power and the curtailment of 5G-BS RE sources, respectively. The overline and underline indicate the maximum and minimum limits of the variable, respectively.

2) Water flow balance

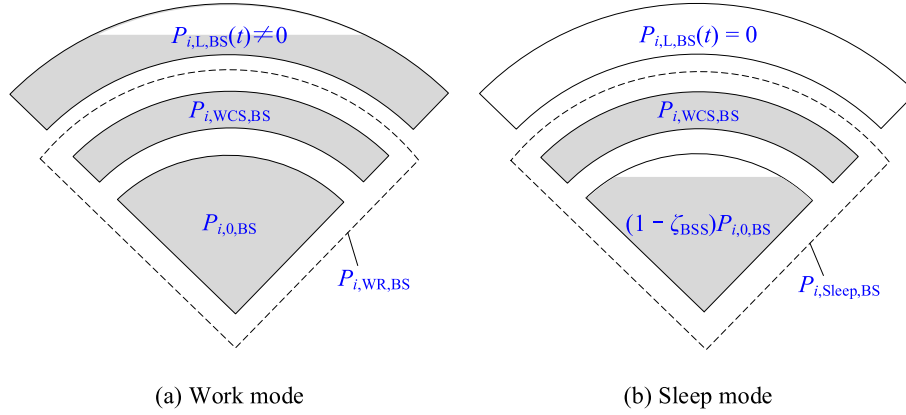


Fig. 3. Power demands of the 5G-BS.

According to Fig. 1 (a), the water balance for the 5G-BS is formulated as

$$q_{m,BS}(t) = q_{m,In,BS}(t) + q_{m,WCS,BS}(t) - q_{m,Out,BS}(t) \quad (7)$$

3) 5G-BS battery

The following constraints represent the operation constraints of the 5G-BS battery:

$$E_{i,Bat,BS}(t) = E_{i,Bat,BS}(t - \Delta t) + \Delta t \left[\eta_{Ch,BS} P_{i,Ch,BS}(t) - \frac{P_{i,Dis,BS}(t)}{\eta_{Dis,BS}} \right] \quad (8)$$

$$\begin{cases} 0 \leq P_{i,Ch,BS}(t) \leq \bar{P}_{i,Ch,BS} \mu_{i,Bat,BS}(t) \\ 0 \leq P_{i,Dis,BS}(t) \leq \bar{P}_{i,Dis,BS} [1 - \mu_{i,Bat,BS}(t)] \end{cases} \quad (9)$$

$$\underline{E}_{i,BS} \leq E_{i,Bat,BS}(t) \leq \bar{E}_{i,BS} \quad (10)$$

$$E_{i,Bat,BS}(T) \geq E_{i,initial,BS} \quad (11)$$

Equation (8) is the energy continuity constraint of the BS battery [43]. The charging/discharging power of the BS battery is restricted by (9), where $\mu_{i,Bat,BS}(t)$ is the 0–1 variable representing the charging/discharging status of the battery, which is equal to 1 if the battery is charging and 0 otherwise. Equation (10) limits the energy of the BS battery in each period. Equation (11) requires that the battery energy at the last time-period T should be no less than the initial value $E_{i,initial,BS}$.

4) 5G-BS water tank

$$H_{m,Ta,BS}(t) = H_{m,Ta,BS}(t - \Delta t) + \frac{\Delta t}{A_{m,Ta,BS}} [q_{m,In,BS}(t) - q_{m,Out,BS}(t)] \quad (12)$$

$$\begin{cases} 0 \leq q_{m,In,BS}(t) \leq \bar{q}_{m,In,BS} \mu_{m,Ta,BS}(t) \\ 0 \leq q_{m,Out,BS}(t) \leq \bar{q}_{m,Out,BS} [1 - \mu_{m,Ta,BS}(t)] \\ M(\mu_{m,Ta,BS}(t) - 1) \leq H_m(t) - H_{m,Ta,BS}(t) \leq M\mu_{m,Ta,BS}(t) \end{cases} \quad (13)$$

$$\underline{H}_{m,Ta,BS} \leq H_{m,Ta,BS}(t) \leq \bar{H}_{m,Ta,BS} \quad (14)$$

$$H_{m,Ta,BS}(T) \geq H_{m,initial,BS} \quad (15)$$

Equation (12) is the water continuity equation for the BS tank. Equation (13) restricts the inflow/outflow rate of the tank, where $\mu_{m,Ta,BS}(t)$ is the 0–1 variable representing the status of the tank, which takes the value of 1 if the tank is filling water and 0 otherwise. Since the 5G-BS tank and the WDS are owned by different service providers, the flow rate of the tank is supposed to be independently controlled through valve controls and thus the BS tank head is decoupled from the nodal head of the WDS. In (13), we use the big-M method to model this control action. For example, $\mu_{m,Ta,BS}(t)$ takes the value of 1 when the nodal head of the WDS $H_m(t)$ is greater than or equal to the BS tank head $H_{m,Ta,BS}(t)$, indicating that water can only flow into the tank at this time, while the flow rate is controllable. Equation (14) limits the head of the BS tank and (15) guarantees that the final tank head at time-period T is no less than initial value $H_{m,initial,BS}$.

3.2. Modeling of the coupled PDS and WDS

3.2.1. Operation constraints of the PDS

A balanced PDS with a radial network topology can be described by the DistFlow model [44], which can be easily extended to unbalanced three-phase power flow equations with some approximations so that the proposed model can also be applied to unbalanced three-phase PDS.

The radial PDS is depicted in Fig. 4, where r_{ij} and x_{ij} are the

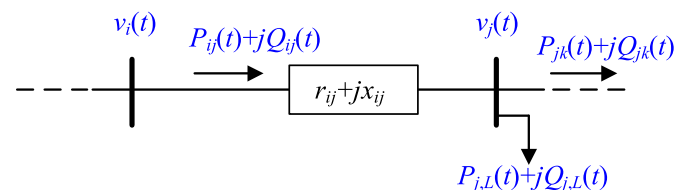


Fig. 4. Schematic diagram of a radial PDS.

resistance and the reactance of power line $i-j$; $v_i(t)$ and $v_j(t)$ are voltage magnitudes of bus i and j , and $P_{ij}(t)$ and $Q_{ij}(t)$ represent the active/reactive power of line $i-j$ flowing from bus i to j , and $P_{j,L}(t)$ and $Q_{j,L}(t)$ are the active/reactive power loads at bus j .

The DistFlow model describes the PDS shown in Fig. 4 with the following active/reactive power and voltage equations:

$$P_{ij}(t) = \sum_{k:j \rightarrow k} P_{jk}(t) + r_{ij} \frac{P_{ij}^2(t) + Q_{ij}^2(t)}{v_i^2(t)} + P_{j,L}(t) \quad (16)$$

$$Q_{ij}(t) = \sum_{k:j \rightarrow k} Q_{jk}(t) + x_{ij} \frac{P_{ij}^2(t) + Q_{ij}^2(t)}{v_i^2(t)} + Q_{j,L}(t) \quad (17)$$

$$v_i^2(t) - v_j^2(t) = 2[r_{ij}P_{ij}(t) + x_{ij}Q_{ij}(t)] - (r_{ij}^2 + x_{ij}^2) \frac{P_{ij}^2(t) + Q_{ij}^2(t)}{v_i^2(t)} \quad (18)$$

Equations (16) and (17) describe the active and reactive power balance of power line $i-j$, where $k:j \rightarrow k$ denotes all the downstream buses connected to bus j , and (18) describes the voltage drop along the power line $i-j$.

In addition, the PDS should operate within the following limits:

$$\underline{P}_{ij} \leq P_{ij}(t) \leq \bar{P}_{ij}, \quad \underline{Q}_{ij} \leq Q_{ij}(t) \leq \bar{Q}_{ij} \quad (19)$$

$$\underline{v}_i \leq v_i(t) \leq \bar{v}_i \quad (20)$$

Note that if bus j in Fig. 4 is connected to PV stations, wind farms, batteries, electrical pumps, etc, installed in the PDS, $P_{j,L}(t)$ and $Q_{j,L}(t)$ should be replaced with the following equivalent loads $P_{j,eq,L}(t)$ and $Q_{j,eq,L}(t)$, respectively,

$$\begin{cases} P_{j,eq,L}(t) = P_{j,L}(t) + P_{j,U}(t) + P_{j,BS}(t) + P_{j,Ch}(t) + P_{j,WGCur}(t) + P_{j,PVCur}(t) \\ \quad - P_{j,G}(t) - P_{j,WG}(t) - P_{j,PV}(t) - P_{j,Dis}(t) \\ Q_{j,eq,L}(t) = Q_{j,L}(t) + Q_{j,U}(t) - Q_{j,G}(t) \end{cases} \quad (21)$$

and the following constraints

$$\underline{P}_{i,G} \leq P_{i,G}(t) \leq \bar{P}_{i,G}, \quad \underline{Q}_{i,G} \leq Q_{i,G}(t) \leq \bar{Q}_{i,G} \quad (22)$$

$$0 \leq P_{j,WGCur}(t) \leq P_{j,WG}(t), \quad 0 \leq P_{j,PVCur}(t) \leq P_{j,PV}(t) \quad (23)$$

$$\begin{cases} 0 \leq P_{i,Ch}(t) \leq \bar{P}_{i,Ch}\mu_{i,Bat}(t) \\ 0 \leq P_{i,Dis}(t) \leq \bar{P}_{i,Dis}[1 - \mu_{i,Bat}(t)] \\ E_{i,Bat}(t) = E_{i,Bat}(t - \Delta t) + \Delta t \left[\eta_{Ch}P_{i,Ch}(t) - \frac{P_{i,Dis}(t)}{\eta_{Dis}} \right] \\ \underline{E}_i \leq E_{i,Bat}(t) \leq \bar{E}_i \\ E_{i,Bat}(T) \geq E_{i,initial} \end{cases} \quad (24)$$

Equation (22) limits the power supplied by power plants, (23) limits the wind and PV curtailment, and (24) includes the operation constraints of the battery installed in the PDS, which are similar to the constraints of the BS battery shown in (8)–(11).

3.2.2. Operation constraints of the WDS

The WDS consists of pipes, tanks, pumps, water sources, and water loads. The operation constraints of WDS include the equality constraints in (25)–(28) and inequality constraints in (29)–(31).

$$\begin{aligned} q_{m,S}(t) + q_{m,Out}(t) - q_{m,In}(t) - q_{m,L}(t) - q_{m,BS}(t) - q_{mn,P}(t) \\ - q_{mn,U}(t) \\ = 0 \end{aligned} \quad (25)$$

$$\Delta H_{mn,P}(t) = H_m(t) - H_n(t) = r_{mn,P} |q_{mn,P}(t)| q_{mn,P}(t) \quad (26)$$

$$\Delta H_{mn,U}(t) = \omega_{mn,U}^2 H_{mn,U0} - A_{mn,U} \omega_{mn,U}^{2-B_{mn,U}}(t) q_{mn,U}^{B_{mn,U}}(t) \quad (27)$$

$$H_m(t) = H_{m,Ta}(t) = H_{m,Ta}(t - \Delta t) + \frac{\Delta t}{A_{m,Ta}} [q_{m,In}(t) - q_{m,Out}(t)] \quad (28)$$

Equation (25) describes the nodal balance at node m , where $q_{mn,P}(t)$ is the flow rate of pipe $m-n$. If water flows from node m to n , $q_{mn,P}(t)$ is positive and negative otherwise. Equation (26) is the Chezy-Manning equation [45] representing the hydraulic characteristics of pipes and (27) describes the head gain of pumps. Equation (28) expresses the water tank.

$$\underline{H}_m \leq H_m(t) \leq \bar{H}_m, \quad \underline{q}_{mn,P} \leq q_{mn,P}(t) \leq \bar{q}_{mn,P} \quad (29)$$

$$\begin{aligned} \underline{H}_{m,Ta} \leq H_{m,Ta}(t) \leq \bar{H}_{m,Ta}, \quad H_{m,Ta}(t) = H_m(t), \quad H_{m,Ta}(T) \geq H_{m,initial}, \\ 0 \leq q_{m,In}(t) \leq \bar{q}_{m,In}\mu_{m,Ta}(t), \quad 0 \leq q_{m,Out}(t) \leq \bar{q}_{m,Out}[1 - \mu_{m,Ta}(t)] \end{aligned} \quad (30)$$

$$\begin{aligned} \Delta \underline{H}_{mn,U} \leq \Delta H_{mn,U}(t) \leq \Delta \bar{H}_{mn,U}, \quad \underline{q}_{mn,U} \leq q_{mn,U}(t) \leq \bar{q}_{mn,U}, \\ \underline{\omega}_{mn,U} \leq \omega_{mn,U}(t) \leq \bar{\omega}_{mn,U} \end{aligned} \quad (31)$$

Equation (29) limits the nodal head and flow rates of pipes, respectively. Equation (30) represents the constraints of tank head and inflow/outflow rates, where $H_{m,Ta}$, representing the head of the tank connected to node m , is assumed to be equal to H_m , which is the head of node m [28]. Equation (31) expresses the boundaries of the flow rate, speed ratio, and head gain of pumps.

3.2.3. Coupling between the PDS and the WDS

The electric pump is the coupling facility between the PDS and the WDS. The active power and reactive power of pumps are determined by (32) and (33) respectively, under the assumption of a constant power factor [24].

$$P_{i,U}(t) = \frac{\rho g}{\eta_{mn,U}} \Delta H_{mn,U}(t) q_{mn,U}(t) \quad (32)$$

$$Q_{i,U}(t) = \gamma_U P_{i,U}(t) \quad (33)$$

3.3. Coordinated operation modes for the IEWDS

The coordinated mechanisms proposed in Section 2.2 are formulated as two optimal problems in this section to define the following two kinds of operation modes.

3.3.1. Operation-mode 1: coordinated operation between the IEWSP and the TSP

Operation-mode 1 realizes the coordinated mechanism presented in Section 2.2.1. Under this mode, the TSP coordinates with the IEWSP by providing the “power-water reserve” service to make a benefit, and accordingly, the IEWSP has the potential to reduce its total operation cost with flexible resources provided by the TSP. Accordingly, operation-mode 1 is formulated as the problem **M1** with the objective function defined by (34)

$$\text{M1: min } C_G + C_{Cur} + C_{Deg} + C_{Bat,BS} + C_{Ta,BS} + C_S \quad (34)$$

and the following constraints:

$$\text{s.t } \begin{cases} 5G - BS \text{ constraints: (1) - (15)} \\ PDS \text{ constraints: (16) - (24)} \\ WDS \text{ constraints: (25) - (31)} \\ \text{Coupling constraints: (32), (33)} \end{cases}$$

As shown in (34), the objective of operation-mode 1 is to minimize the overall operation cost of the IEWSP, which includes 1) the cost for purchasing electricity and water, denoted by C_G and C_S , respectively 2) the cost of curtailing RE sources of the IEWSP, denoted by C_{Cur} , 3) the battery operation cost C_{Deg} , and 4) the compensation of the battery service and tank service to the TSP, denoted by $C_{Bat,BS}$ and $C_{Ta,BS}$, respectively. All these costs are defined by

$$\begin{cases} C_G = \sum_{t=1}^T C_{1,G}(t) P_{1,G}(t) \\ C_{Cur} = \sum_i \sum_{t=1}^T [C_{i,WGCur} P_{i,WGCur}(t) + C_{i,PVCur} P_{i,PVCur}(t)] \quad i \in RES \\ C_{Deg} = \sum_i C_{i,Deg} \sum_{t=1}^T [P_{i,Ch}(t) + P_{i,Dis}(t)] \quad i \in RE \\ C_{Bat,BS} = \sum_i C_{i,Bat,BS} \sum_{t=1}^T [P_{i,Ch,BS}(t) + P_{i,Dis,BS}(t)] \quad i \in BS \\ C_{Ta,BS} = \sum_i C_{m,Ta,BS} \sum_{t=1}^T [q_{m,In,BS}(t) + q_{m,Out,BS}(t)] \quad i \in BS \\ C_S = \sum_{t=1}^T C_{1,S}(t) q_{1,S}(t) \end{cases} \quad (35)$$

Note that detailed calculation steps for the battery degradation cost $C_{i,Deg}$ are provided in Refs. [43,46]. The variables $C_{i,Bat,BS}$ and $C_{m,Ta,BS}$ are the compensation-coefficients for the BS battery service and the BS tank service, respectively. The case where $C_{i,Bat,BS}$ takes the smallest value is to only compensate for the BS battery degradation cost, in order to compensate the TSP's battery maintenance

cost. The value of $C_{m,Ta,BS}$ is related to multiple factors such as the deployment cost in different areas and facility aging [47].

3.3.2. Operation-mode 2: dual coordinated operations among the IEWSP, the TSP, and the upper-level grid

Operation-mode 2 realizes the coordinated mechanism presented in Section 2.2.2. Under this mode, coordination between the IEWSP and the upper-level grid is also introduced besides the coordination between the IEWSP and the TSP. The IEWSP is supposed to sign a contract with the upper-level grid. The contract stipulates that the upper-level grid sells low-priced electricity to the IEWSP all day, which acts as the compensation for the IEWSP to track the desired tie-line power curve. However, the IEWSP has to pay for the power deviation.

Operation-mode 2 is formulated as the problem **M2** with the objective function defined by

$$\begin{aligned} \text{M2: min } & C_{Cut} + C_G^* + C_{Cur} + C_{Deg} + C_{Bat,BS} + C_{Ta,BS} + C_S \\ C_{Cut} = & C_{1,Cut} \sum_{t=1}^T |P_{1,G}(t) - P_{1,Pre}(t)| \\ C_G^* = & \sum_{t=1}^T C_{1,G}^*(t) P_{1,G}(t) \end{aligned} \quad (36)$$

and the same constraints of problem **M1**.

Note that the overall cost for the power deviation depends on the coefficient $C_{1,Cut}$ multiplied by the absolute value of the difference between power $P_{1,G}(t)$ and the desired power $P_{1,Pre}(t)$, which can be simplified by Ref. [48]. The variable C_G^* denotes the new power purchase cost.

4. Solution methodology

Both **M1** and **M2** formulated in Section 3.3 are intractable MINLP problems due to the plenty of nonlinear functions involved. In this section, the original problem will be reformulated into a MISOCP problem with the techniques of SOCP relaxation and piecewise linearization.

4.1. SOCP relaxation for PDS operation constraints

In this work, the nonlinear PDS functions are addressed through the SOCP relaxation in Ref. [49]. The tightness and exactness of the SOCP relaxation have been proved in Ref. [50].

Two auxiliary variables, $l_{ij}(t)$ and $u_i(t)$ are defined to substitute the nonlinear terms.

$$l_{ij}(t) = \frac{P_{ij}^2(t) + Q_{ij}^2(t)}{v_i^2(t)} \quad (37)$$

$$u_i(t) = v_i^2(t) \quad (38)$$

Then the PDS constraints in (16)–(18) are rewritten as:

$$P_{ij}(t) = \sum_{k:j \rightarrow k} P_{jk}(t) + r_{ij} l_{ij} + P_{j,L}(t) \quad (39)$$

$$Q_{ij}(t) = \sum_{k:j \rightarrow k} Q_{jk}(t) + x_{ij} l_{ij} + Q_{j,L}(t) \quad (40)$$

$$u_i(t) - u_j(t) = 2[r_{ij} P_{ij}(t) + x_{ij} Q_{ij}(t)] - (r_{ij}^2 + x_{ij}^2) l_{ij} \quad (41)$$

$$l_{ij}(t) \geq \frac{P_{ij}^2(t) + Q_{ij}^2(t)}{u_i(t)} \quad (42)$$

Obviously, (42) can be rearranged to a second-order cone constraint.

4.2. Linearization of WDS operation constraints

The piecewise linearization technique is applied to approximate the nonlinear functions (26), (27) and (32). The case studies in Ref. [27] indicate that the technique provides acceptable solution accuracy at a reasonable computational time.

4.2.1. Linearization of the head loss function

For (26), the range of $q_{mn,p}(t)$ is firstly divided into $S-1$ intervals by S breakpoints, as shown in Fig. A1(a) in Appendix A. Then S continuous variables $\lambda_{mn,s}(t)$ ($\lambda_{mn,s}(t) \in [0,1]$, $s = 1, 2, \dots, S$) are introduced and related to corresponding breakpoints of $q_{mn,p}(t)$, and $S-1$ binary variables $a_{mn,s}(t)$ ($s = 1, 2, 3, \dots, S-1$) are related to corresponding intervals divided by the breakpoint.

Therefore, the variable $q_{mn,p}(t)$ and equation (26) can be rewritten as (43) and (44), respectively [27].

$$q_{mn,p}(t) = \sum_{s=1}^S [\lambda_{mn,s}(t) q_{mn,p,s}] \quad (43)$$

$$\Delta \hat{H}_{mn,p}(t) = \sum_{s=1}^S \left\{ \lambda_{mn,s}(t) \Delta H_{mn,p} [q_{mn,p,s}] \right\} \quad (44)$$

$$\sum_{s=1}^{S-1} a_{mn,s}(t) = 1 \quad (45)$$

$$\sum_{s=1}^S \lambda_{mn,s}(t) = 1 \quad (46)$$

$$\lambda_{mn,1}(t) \leq a_{mn,1}(t) \quad (47)$$

$$\lambda_{mn,s}(t) \leq a_{mn,s-1}(t) + a_{mn,s}(t) \quad s = 2, 3, \dots, S-1 \quad (48)$$

$$\lambda_{mn,S}(t) \leq a_{mn,S-1}(t) \quad (49)$$

4.2.2. Linearization of the water pump constraints

For the bivariate functions (27) and (32), as shown in Fig. A1(b) in Appendix A, the range of $q_{mn,u}(t)$ and $\omega_{mn,u}(t)$ are divided into $X-1$ and $Y-1$ intervals by X and Y breakpoints, respectively. The four coordinated points are the vertices of the rectangle which is divided into an upper triangle and a lower one. Continuous variables $\lambda_{mn,x,y}(t)$ ($\lambda_{mn,x,y}(t) \in [0,1]$, $x = 1, 2, \dots, X$, $y = 1, 2, \dots, Y$), and binary variables $a_{up,mn,x,y}(t)$ and $a_{do,mn,x,y}(t)$ are introduced. Each continuous variable is associated with a corresponding vertex, and each binary variable is associated with a triangle, as shown in Fig. A1(b).

Therefore, $q_{mn,u}(t)$ and $\omega_{mn,u}(t)$ are rewritten as (50) and (51), and the approximation of (27) and (32) are shown in (52) and (53), respectively [27].

$$q_{mn,u}(t) = \sum_{x=1}^X \sum_{y=1}^Y [\lambda_{mn,x,y}(t) q_{mn,u,x}] \quad (50)$$

$$\omega_{mn,u}(t) = \sum_{x=1}^X \sum_{y=1}^Y [\lambda_{mn,x,y}(t) \omega_{mn,u,y}] \quad (51)$$

$$\Delta \hat{H}_{mn,u}(t) = \sum_{x=1}^X \sum_{y=1}^Y \left\{ \lambda_{mn,x,y}(t) \Delta H_{mn,u} [q_{mn,u,x}, \omega_{mn,u,y}] \right\} \quad (52)$$

$$\hat{P}_{mn,u}(t) = \sum_{x=1}^X \sum_{y=1}^Y \left\{ \lambda_{mn,x,y}(t) P_{mn,u} [q_{mn,u,x}, \omega_{mn,u,y}] \right\} \quad (53)$$

$$\sum_{x=1}^{X-1} \sum_{y=1}^{Y-1} [a_{up,mn,x,y}(t) + a_{do,mn,x,y}(t)] = 1 \quad (54)$$

$$\sum_{x=1}^X \sum_{y=1}^Y \lambda_{mn,x,y}(t) = 1 \quad (55)$$

$$\begin{aligned} \lambda_{mn,x,y}(t) &\leq a_{up,mn,x,y-1}(t) + a_{up,mn,x+1,y}(t) + a_{up,mn,x,y}(t) \\ &\quad + a_{do,mn,x-1,y}(t) + a_{do,mn,x,y+1}(t) + a_{do,mn,x,y}(t) \end{aligned} \quad (56)$$

4.3. Reformulated model

With the SOCP relaxation and linearization above, the original problem M1 and M2 are reformulated as M1* and M2*, with the objectives defined by

$$\begin{cases} \text{M1}^*: \min C_G + C_{Cur} + C_{Deg} + C_{Bat,BS} + C_{Ta,BS} + C_S \\ \text{M2}^*: \min C_{Cut} + C_G^* + C_{Cur} + C_{Deg} + C_{Bat,BS} + C_{Ta,BS} + C_S \end{cases} \quad (57)$$

and the same constraints:

$$s.t. \begin{cases} 5G-BS \text{ constraints: (1) - (15)} \\ PDS \text{ constraints: (19) - (24), (37) - (42)} \\ WDS \text{ constraints: (25), (28) - (31), (43) - (52), (54) - (56)} \\ \text{Coupling constraints: (53), (33)} \end{cases}$$

To provide a quick reference, the decision variables, objectives, and constraints of the optimal models proposed are summarized in Fig. 5.

5. Case studies

5.1. Test system description and data sources

The proposed model is applied to two test IEWDSs. System 1 is composed of the IEEE-33 PDS [44] and an 8-node WDS [46], and System 2 is a larger test system composed of the IEEE-123 PDS [51] and a 24-node WDS. Simulations in Section 5.2 and 5.3 are conducted on System 1 to illustrate the proposed operation modes and discuss the effects of the BS battery capacity and compensation of the BS battery service, respectively. Simulations in Section 5.4 are conducted on System 2 to verify the adaptability of the proposed

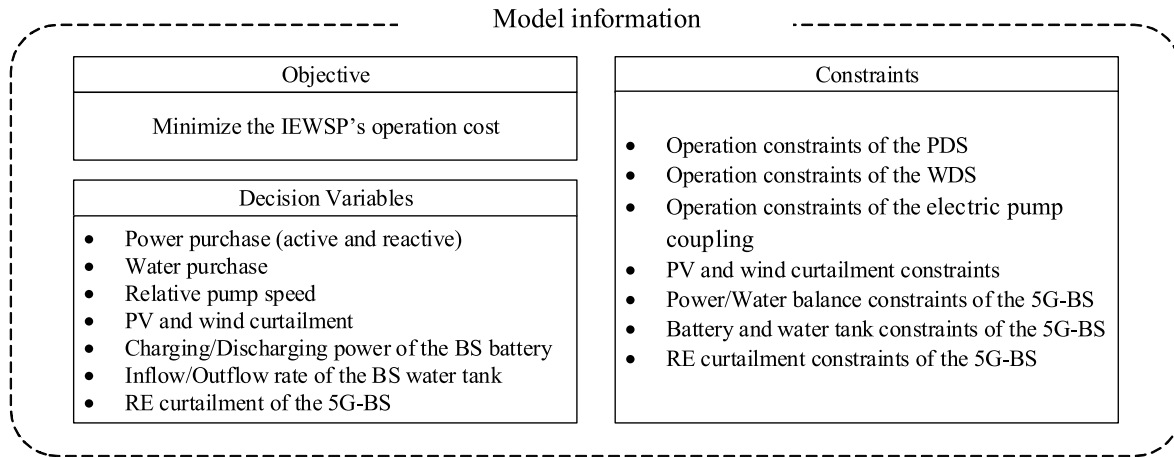


Fig. 5. Summary of the proposed model.

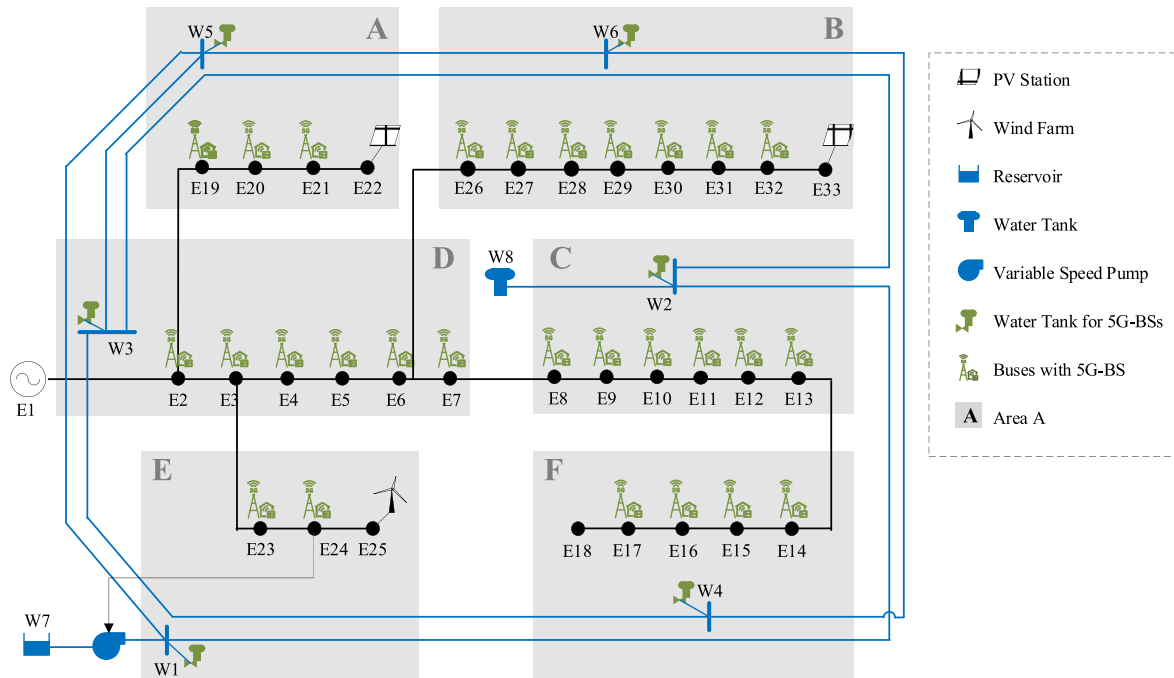


Fig. 6. Topology of system 1.

Table 1
PDS buses and WDS nodes in each area.

Area	PDS Buses	WDS Nodes	Buses connected to 5G BS	Nodes with a BS water tank
A	E19–E22	W5	E19–E21	W5
B	E26–E33	W6	E26–E32	W6
C	E8–E13	W2	E8–E13	W2
D	E2–E7	W3	E2–E7	W3
E	E23–E25	W1	E23, E24	W1
F	E14–E18	W4	E14–E17	W4

model to a larger system.

System 1 is shown in Fig. 6, where Letter E and W denote PDS buses and WDS nodes, respectively. The test system is divided into six areas denoted by the gray rectangles in Fig. 6. Considering that the service coverage radius of a 5G-BS is generally between 100 and

300 m [10], almost all the PDS buses are assumed to connect to a 5G-BS. In addition, all the 5G-BSs in the test system are supposed to be water-cooled and a BS water tank connected to the WDS is installed in each area to supply water to the BSs within the corresponding area. BSs and BS tanks in each area and their connections

Table 2
Electricity and water prices.

Energy price	22:00–8:00	8:00–22:00
TOU electricity price (\$/kWh)	0.0374	0.1195
Contract electricity price (\$/kWh)	0.0374	
Contract deviation electricity price (\$/kWh)	0.0821	
Comprehensive water price (\$/m ³)	0.8269	

to the PDS and the WDS are shown in Table 1.

The total power load [52], the total 5G-BS load [53], the sum of the first two, and the total water load [54] of the test system during a one-day horizon are summarized in Fig. A2(a). The total wind power output [52], the total PV output [52], and the total RE output of 5G-BSs are shown in Fig. A2(b).

Table 2 shows the TOU electricity price and the comprehensive water price, which are based on the typical price data of cities in China. Note that the contract electricity price is equal to the valley electricity price and remains unchanged throughout the day. However, the active power deviation will be charged according to the contract deviation electricity price.

All the parameters and data for this simulation can be found in Ref. [46].

5.2. Two coordinated operation modes

The four cases as shown in Table 3 are developed to illustrate the two proposed cooperation mechanisms between the TSP and the IEWSP.

5.2.1. Operation-mode 1

The effectiveness of operation-mode 1 is investigated in this subsection by comparing the results of Case 1 and Case 2. In Case 1, the IEWSP solves an optimal operation problem to minimize its operation cost without considering the TSP's service. It is supposed that the BS battery is only used for the emergency power supply of the 5G-BS and does not participate in the daily dispatch. The 5G-BS mainly obtains power from the PDS and its on-site RE source. In Case 2, the IEWDS and the 5G-BSs are operated in a coordinated way. The TSP provides the BS battery and the regional BS water tank as dispatchable resources to the IEWSP, and the corresponding operation costs for the dispatchable resources are compensated by the IEWSP with a contracted compensation coefficient.

The daily operation costs of the IEWSP and the compensations to the TSP in Cases 1 and 2 are shown in Table 4. It can be observed that both the IEWSP and the TSP can benefit from the coordinated operation. Compared with Case 1, the IEWSP's operation cost is reduced by \$103.08, and the TSP receives an additional compensation of \$184.94 in Case 2.

The tie-line powers in Case 1 and Case 2 over the 24 h are depicted in Fig. 7. In Case 2, an obvious increase of tie-line power during the valley price period (22:00–8:00), as well as a decrease

during the peak price period (8:00–22:00), can be observed. This is because in Case 2, the battery and water tanks are dispatchable to the IEWSP owing to the coordinated operation. The IEWSP in Case 2 can choose to purchase power from the external grids or to compensate the power supply by the TSP, and therefore achieve a reduction in the electricity bill.

Besides, the proposed operation-mode 1 used in Case 2 is also helpful to accommodate the surplus RE powers in the 5G-BS. There is no other way for the TSP to accommodate its surplus RE powers but to store in the BS batteries in Case 1, which results in the RE curtailment of 627.7 kW. While in Case 2, the surplus RE powers are allowed to transmit to the IEWDS, and it is found that all the RE powers are accommodated and there is no RE curtailment in both the IEWDS and the 5G-BSs.

To analyze the role of the BS battery and the regional BS water tank throughout the day, more results are shown in Figs. 8–10. The total charging/discharging power for all the BS batteries in Case 2 is shown in Fig. 8, where a minus power indicating discharging. The pump power in Case 1 and Case 2, as well as the total inflow/outflow rate for all the BS water tanks in Case 2, are shown in Figs. 9 and 10, respectively.

As shown in Fig. 8, the BS batteries are in discharging states during the peak price period (8:00–22:00) and in charging states during the valley price period (22:00–8:00). That is, the BS batteries act as power sources to the IEWDS during 8:00–22:00 to reduce the power purchased by the IEWSP from the external grid with a high cost. For example, at time-period 19 when the power load of the IEWDS is high (see Fig. A2(a)), it is found that the discharging power of the BS batteries reaches the peak value of 0.45 MW, accounts for 12.82% of the tie-line power at the same time. While after the time-period 22, the BS batteries begin to be charged with low-cost electricity and guarantee that their energy stored at the end of the day is not less than the initial value.

Meanwhile, compared with Case 1, the variable-speed pump of the IEWDS in Case 2 can make a better speed-adjustment strategy to reduce power consumption owing to the additional water storage capacity provided by the BS tanks. For example, during the time-period 2–3 and 22–24 (the valley price period), the pump power in Case 2 increases (see Fig. 9), and the inflow rate of the BS tanks increases accordingly (see Fig. 10), which indicates that the pump in Case 2 pressurizes the WDS and charges the BS tanks with the valley-priced electricity. On the contrary, during the time-period 8 and 18–21, although the electricity is at a high price, the pump in Case 1 has to consume much electricity to satisfy the high water demand (see Fig. A2(a)); while in Case 2, since some water load is fed by releasing the water storage from the BS tanks (see Fig. 10), a reduction of pump power, and accordingly of electricity bill, is achieved. Note that the large change in the BS tank outflow rate at time-period 19 is due to the increase in pump power to cope with the increased water load at the period, so water flows into the tank.

Table 3
The four cases developed.

Case	Interaction between TSP and IEWSP considered	Interaction between IEWSP and the upper-level grid considered	Description
Case 1	No	No	The IEWSP operates its PDS and WDS separately without coordinating with the TSP or the upper-level grid.
Case 2	Yes	No	The IEWSP operates under operation-mode 1 , i.e., the IEWSP operates its PDS and WDS considering only the coordination with the TSP.
Case 3	No	Yes	Only the coordination between the IEWSP and the upper-level grid is considered.
Case 4	Yes	Yes	The IEWSP operates under operation-mode 2 , i.e., dual coordinated operations among the IEWSP, the TSP, and the upper-level grid are considered.

Table 4
Operation costs and the compensation in Case 1 and Case 2 (operation-mode 1).

Case	IEWSP's operation cost (\$)	TSP's compensation (\$)
Case 1	16928.93	0
Case 2 (operation-mode 1)	16825.85	184.94
Increment	-103.08	184.94

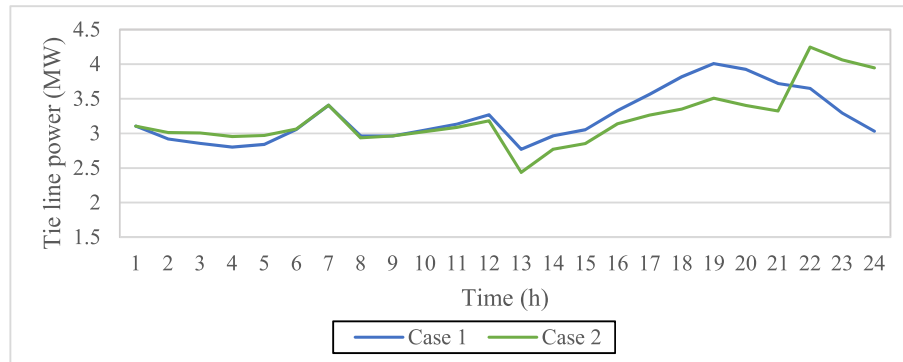


Fig. 7. The tie-line (E1-E2) power in Case 1 and Case 2.

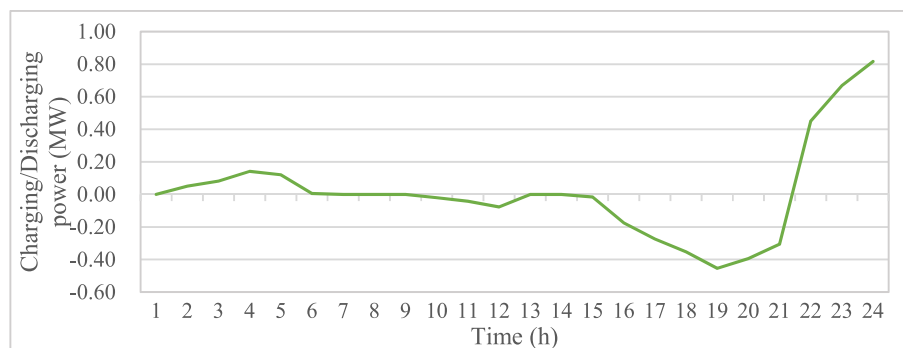


Fig. 8. Total charging/discharging powers of all 5G-BS batteries in Case 2.

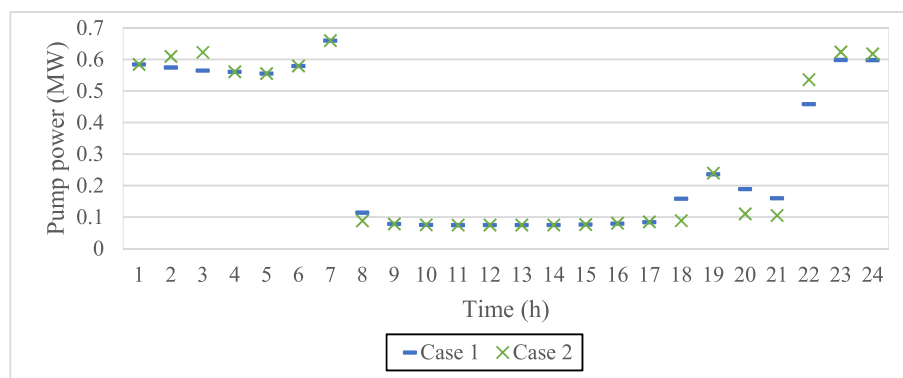


Fig. 9. The pump power in Case 1 and Case 2.

5.2.2. Operation-mode 2

In this subsection, operation-mode 2, where the coordination mechanism between the IEWSP and the upper-level grid is incorporated, is studied with Case 3 and Case 4.

The operation costs of the IEWSP and the compensations to the TSP in Cases 3 and 4 are shown in Table 5. It can be observed that both the IEWSP and the TSP can benefit from the coordination of

operation-mode 2, which is similar to the observation from the results of operation-mode 1 (see Table 4). Compared with Case 3, the IEWSP's operation cost is reduced by \$70.12, and the TSP receives additional compensation of \$14.90 in Case 4.

The tie-line powers in Case 3 and Case 4, as well as the desired one provided by the upper-level grid, are depicted in Fig. 11. It is shown that the tie-line power in Case 4 follows the desired curve

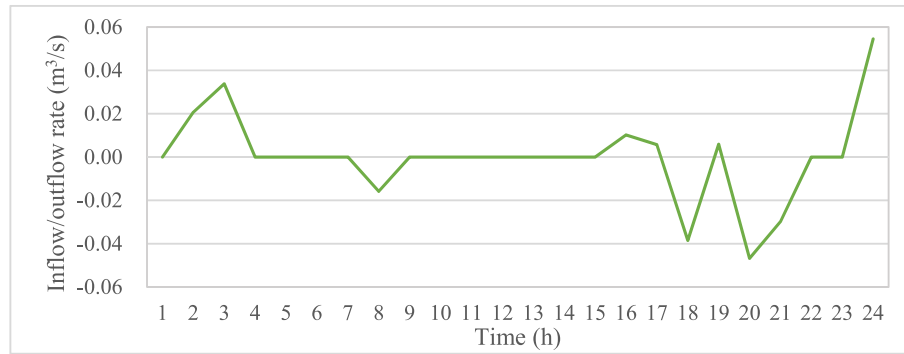


Fig. 10. Total inflow/outflow rates of all 5G-BS tanks in Case 2.

Table 5

Operation costs and the compensation in Case 3 and Case 4.

Case	IEWSP's operation cost (\$)	TSP's compensation (\$)
Case 3	13171.35	0
Case 4 (operation mode 2)	13101.23	14.90
Increment	-70.12	14.90

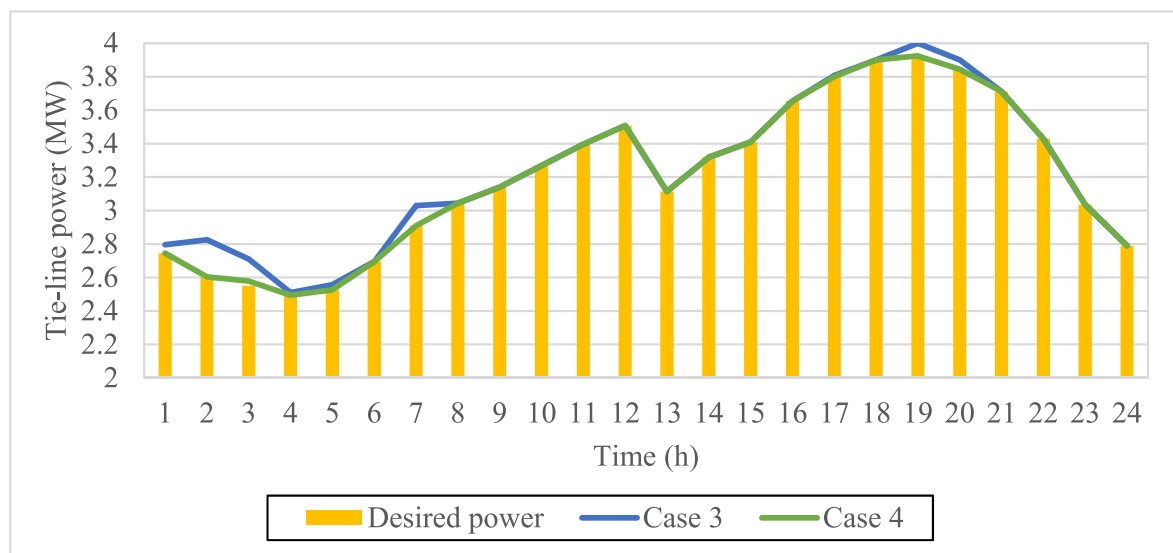


Fig. 11. The tie-line power (E1-E2) profile in Case 3 and Case 4 (operation-mode 2).

well while obvious deviations in Case 3 can be observed. For example, the power deviation in Case 3 at time-period 2 reaches up to 0.22 MW, which accounts for 7.86% of the tie-line power at the same period.

The IEWSP in Case 3 pays \$51.64 to the upper-level grid due to the power deviation according to the contract. In contrast, with the help of the dispatchable resources provided by the TSP, the IEWSP in Case 4 can control the tie-line power to follow the desired curve as well as possible. Since the compensation paid to the TSP is less than the punishment for the power deviation charged by the upper-level grid, the IEWSP's operation cost in Case 4 is reduced compared with Case 3.

5.3. Sensitivity analysis

In this section, two sensitivity studies are conducted based on Case 2 to evaluate the effects of the BS battery capacity and

Table 6

The operation cost and the compensation with different 5G-BS battery capacities.

Scene	Capacity (kWh)	IEWSP's operation cost (\$)		TSP's compensation (\$)	
		Value	Increment	Value	Increment
1	200	16825.85	—	184.94	—
2	400	16824.92	-0.93	212.25	27.31
3	600	16824.26	-1.59	229.81	44.87
4	800	16824.25	-1.60	239.79	54.85

compensation-coefficient of the BS battery service, respectively, on the coordination between the IEWSP and the TSP.

5.3.1. BS battery capacity

The simulation in Case 2 is conducted with the 5G-BS battery capacities (in a single 5G-BS) being 200 kWh, 400 kWh, 600 kWh,

and 800 kWh, which are denoted as Scene 1–4, respectively. The resulting operation cost and the compensation are listed in Table 6.

The following observations can be made from Table 6.

- (1) Compared with Scene 1, the IEWSP's operation cost experiences a slight drop as the BS battery capacity increases, indicating that the increase in the BS battery capacity has little effect on reducing the operation cost of the IEWSP. For example, as the battery capacity increase to 800 kWh, only a reduction of \$1.6 in operation cost can be achieved.
- (2) In contrast, the TSP's compensation increases obviously with the increase in the BS battery capacity. For example, the TSP's compensation in Scene 4 is \$54.85 more than that of Scene 1, which accounts for 29.66% of the TSP's compensation in Scene 1.

The total charging/discharging power of all 5G-BS batteries with different capacities is shown in Fig. 12. It can be found that during the time-period 1–7 (the valley price period), the 5G-BS batteries in Scene 1–4 are in charging status with a successively increased charging power as the capacity increases from 200 kWh to 800 kWh. In contrast, during the time-period 8–21 (the peak price period), the 5G-BS batteries in Scene 1–4 are in discharging status with a successively increased discharging power as the capacity increases from 200 kWh to 800 kWh. It is interesting to find that during the time-period 22–24 (the valley price period), as the battery capacity increases, the charging power decreases. This is because the energy stored in the batteries at the end of the day should be no less than their initial value (see (11)).

Results of Table 6 and Fig. 12 demonstrate that the 5G-BS battery capacity has a great influence on the TSP's compensation, but little influence on the IEWSP's operation cost. As the battery capacity increases, the TSP can provide more power support to the IEWSP and therefore obtain more compensation, while for the IEWSP, the battery compensation cost increases (see the objective in (34)), resulting in only a slight reduction in the operation cost.

5.3.2. Compensation-coefficient for the BS battery service

The simulation in Case 2 is conducted with the compensation-coefficient for the BS battery service being decreased and increased by 10% based on that in Scene 1, which are denoted as Scene 5 and 6, respectively. The resulting operation cost and the compensation are listed in Table 7.

The following observations can be made from Table 7.

- (1) Compared with Scene 1, the IEWSP's operation cost increases slightly with the increase of the compensation-coefficient and vice versa. For example, as the compensation-coefficient increases by 10% and reaches 0.0434 \$/kWh, an increment of \$15.43 in operation cost can be observed; while the IEWSP's operation cost drops by \$22.32 as the compensation-coefficient decreases by 10% and reaches 0.0355 \$/kWh.
- (2) In contrast, the TSP's compensation experiences a drastic reduction and a substantial increase with the increase and decrease in the compensation-coefficient, respectively. For example, the TSP's compensation in Scene 5 is \$186.13 more than that of Scene 1, which accounts for 100.65% of the TSP's compensation in Scene 1. On the contrary, the TSP's compensation in Scene 6 is \$166.59 less than that of Scene 1, which accounts for 90.08% of the TSP's compensation in Scene 1. These results indicate that the variation in the compensation-coefficient has a greater effect on the benefit of the TSP.

The total charging/discharging power of all the 5G-BS batteries with different compensation-coefficients is shown in Fig. 13. It can be observed that the IEWSP significantly increases the use of the BS battery during the time-period 16–21 (the peak price period). In this period, the total power load in the PDS reaches the peak of the day (see Fig. A2(a)). Due to the reduced compensation-coefficient cost of the TSP in Scene 5, the IEWSP prefers to use the 5G-BS battery with a moderate compensation rather than using the high-

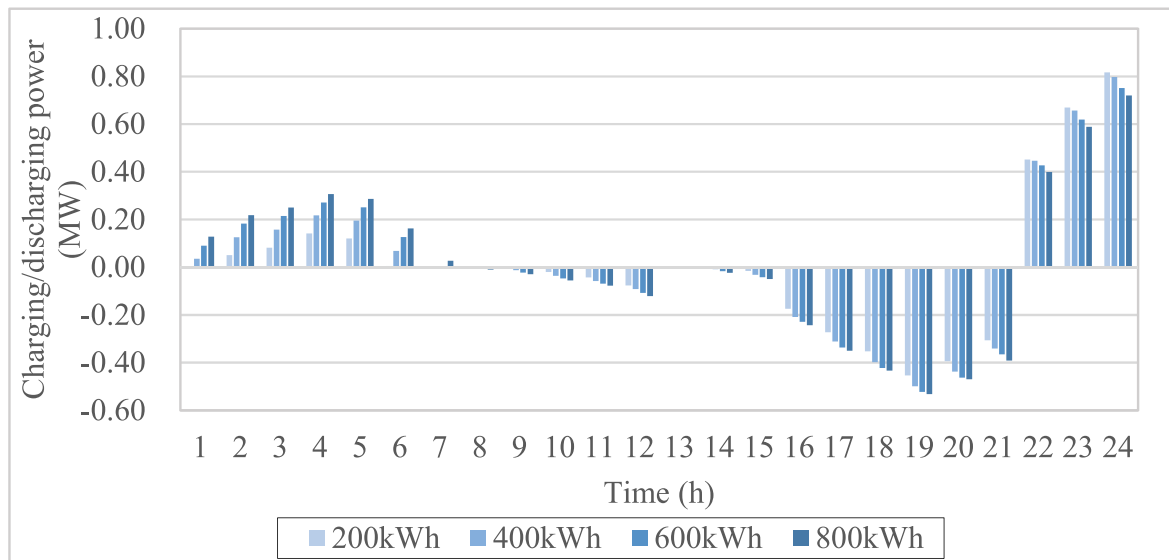
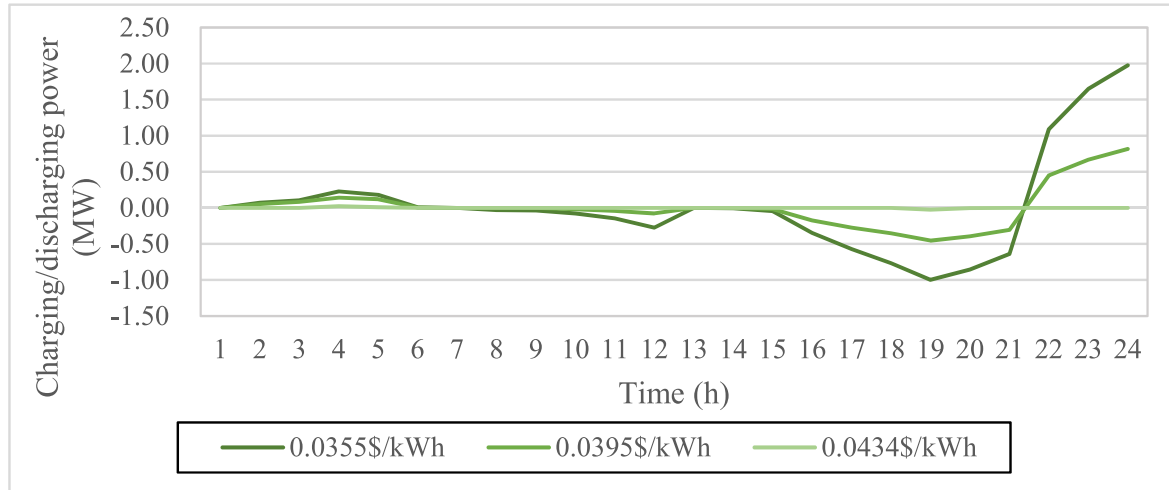


Fig. 12. Total charging/discharging powers of all 5G-BS batteries with different capacities.

Table 7

Operation costs and compensations with different compensation-coefficients for battery service.

Scene	Coefficient (\$/kWh)	Coefficient increase ratio	IEWSP's operation cost (\$)		TSP's compensation (\$)	
			Value	Increment	Value	Increment
1	0.0395	—	16825.85	—	184.94	—
5	0.0355	−10%	16803.53	−22.32	371.07	186.13
6	0.0434	10%	16841.28	15.43	18.35	−166.59

**Fig. 13.** Charging/discharging powers of all the 5G-BS batteries with different compensation-coefficients.

price electricity from the grid. Therefore, the discharging power of the 5G-BS batteries climbs up to 1 MW, which is 122.22% higher than that of Scene 1 for the same period. After the time-period 22 (the valley price period), the 5G-BS battery is charged with a maximum charging power of 1.98 MW, which is 141.46% higher than that of Scene 1 in this period. Conversely, as the compensation-coefficient of the TSP increases (Scene 6), the IEWSP greatly reduces the use of the 5G-BS battery and the entire charging/discharging curve tends to be flat.

Results of Table 7 and Fig. 13 show that the compensation-coefficient of the 5G-BS battery affects both the IEWSP's operation cost and the TSP's compensation. With the development of battery techniques, the TSP can provide more battery services with reduced cost and both the TSP and the IEWSP will make greater benefits from the coordination. On the contrary, with the increase in battery compensation cost, the coordination benefits for the TSP and the IEWSP will reduce. If the compensation cost for the 5G-BS battery exceeds some value, the IEWSP will no longer use this battery service, which indicates that there is no coordination between the TSP and the IEWSP.

5.4. Adaptability to a larger test system

In this section, Cases 1–4 presented in Section 5.2 are restudied with System 2, which is a larger IEWDS composed of the IEEE-123 PDS and a 24-node WDS.

The IEWSP's operation costs, the TSP's compensations, and the electricity and water purchasing in the four cases of System 2 are summarized in Table 8.

Comparing the results of Case 1 and Case 2 in Table 8 leads to a similar conclusion as in Section 5.2.1, and comparing Case 3 and Case 4 leads to a similar conclusion as in Section 5.2.2. That is, under both operation modes, the coordinated operation (Cases 2 and 4) will bring benefit to the two sides of the IEWSP and the TSP.

A comparison of the two operation modes will be made further in this section. Comparing the non-coordination cases, i.e. Case 1 and Case 3, of the two modes, it is found that the IEWSP's operation cost in Case 3 is reduced by \$4008.86 compared with that in Case 1. The reduction accounts for 10.54% of the operation cost in Case 1. Comparing the coordinated cases of Cases 2 and 4, it is obtained that the IEWSP's operation cost in Case 4 is reduced by \$3605.57 compared with that in Case 2, which accounts for 9.61% of the

Table 8

Operation costs, compensations, and costs for electricity and water purchasing of System 2.

Case	IEWSP's operation cost (\$)	TSP's compensation (\$)	Electricity purchased (MWh)	Water purchased (m ³)
1	38035.85	0	87.76	37043.04
2	37523.98	184.98	85.38	37061.11
3	34026.99	0	87.89	37043.04
4	33918.41	12.91	86.54	37076.01

operation cost in Case 2. These results demonstrate that the IEWSP can benefit from following the power curve given by the upper-level grid.

It is also shown by Table 8 that to follow the power curve, the IEWSP should purchase a little more electricity and water in Cases 3/4 compared with Case 1/2. For example, the amount of electricity purchased in Cases 3 is 0.13 MWh more than that of Cases 1, and the water purchased in Case 4 is 14.9 m³ more than that of Case 2. However, since the increment in cost for purchasing electricity and water in Case 3 and Case 4 is small, the IEWSP's total operation costs in Case 3/4 decrease compared with those of Case 1/2.

As for the TSP, it is shown that the compensation in Case 2 is \$172.07 more than that of Case 4, indicating that the IEWSP reduces the utilization of the service provided by the TSP in Case 4. This is because the IEWSP in operation-mode 2 can enjoy the valley electricity price throughout the day according to the contact. Therefore, the IEWSP can make an optimal decision to minimize its own operation cost by choosing to follow the desired power of the upper-level grid with a reward of the valley electricity price or to use the service of battery and water tank provided by the TSP with the deviation penalty cost. In this simulation, the valley electricity price is set to be much lower than the compensation of the 5G-BS battery service, and the penalty for power deviation is high. Therefore, the IEWSP prefers to follow the power curve strictly, which may reduce the compensation for the TSP.

6. Conclusion

This paper highlights the interaction between the 5G-BSs and the IEWDS and investigates the possible win-win cooperation mechanism between the TSP and IEWSP.

An optimal model for the coordinated operation with detailed

operation constraints of the 5G-BS and the IEWDS is proposed and two modes for the operation of the IEWDS coordinating with the TSP and the upper-level grid are further developed. The proposed model is reformulated into a tractable MISOCP problem with the techniques of SOCP relaxation and piecewise linearization.

Case studies are carried out with two test IEWDSs. Four cases under the two operation modes are simulated and compared, as well as sensitivity and adaptability studies are conducted. Simulation results demonstrated the effectiveness of the proposed optimal model and its solution method, and more importantly, recognized the mutual benefits for both the TSP and the IEWSP from the coordinated operation between the IEWDS and 5G-BSs.

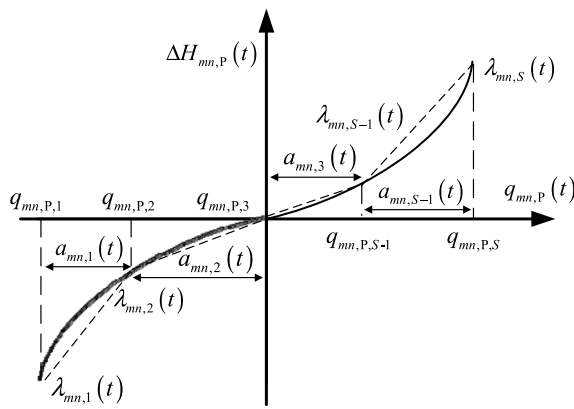
Credit author statement

Mingyi Sun: Conceptualization, Methodology, Software, Writing original draft, Investigation, Validation, Data curation. **Xia Zhao:** Methodology, Supervision, Writing Reviewing and Editing, Visualization. **Hong Tan:** Writing Reviewing and Editing. **Xinyi Li:** Writing Reviewing and Editing.

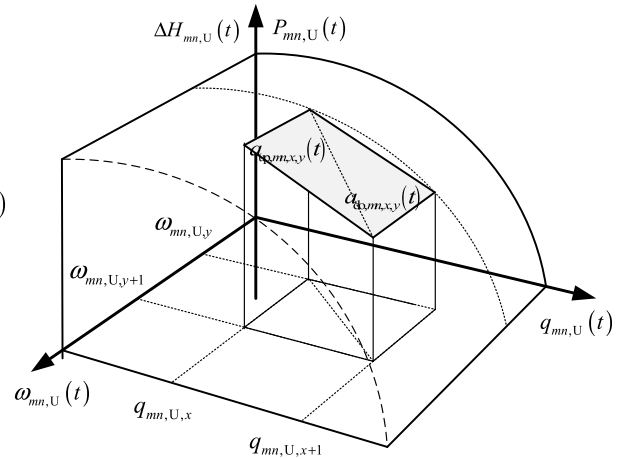
Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A

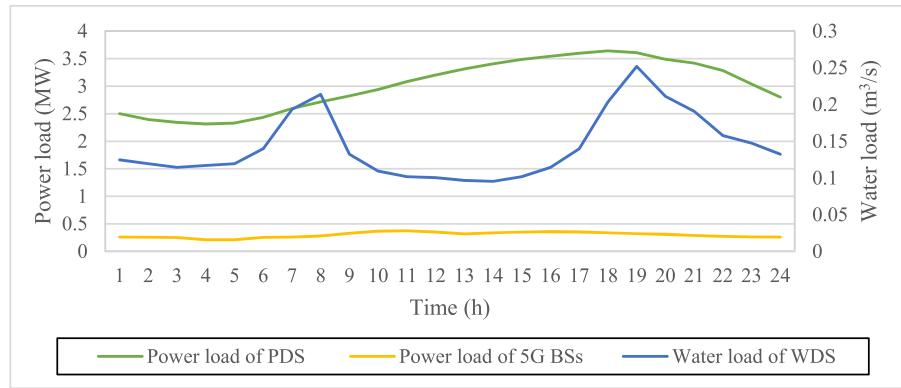


(a) hydraulic head loss function

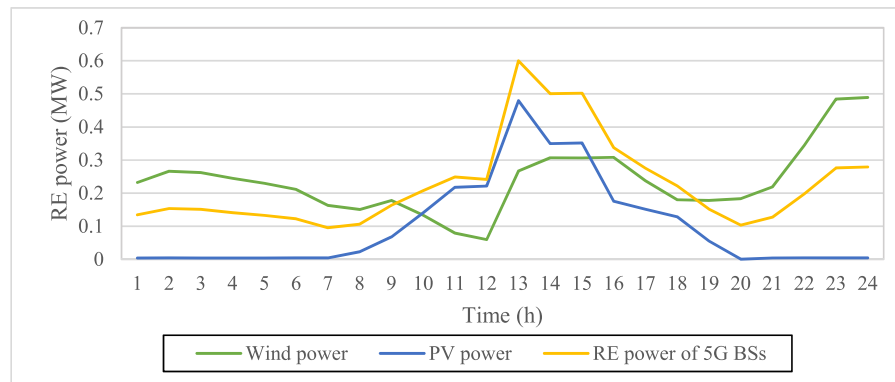


(b) water pump functions

Fig. A1. Piecewise linearization of the hydraulic head loss function and the water pump functions



(a) Load demand



(b) RE power outputs

Fig. A2. Load demand and RE power output curves

Tabel A1

Bounds on the decision variables

Decision variables	Upper limit	Lower limit	Units
$P_{i,Ch,BS}(t)$	0.1	0	MW
$P_{i,Dis,BS}(t)$	0.1	0	MW
$\mu_{i,Bat,BS}(t)$	1	0	0-1 variable
$q_{m,in,BS}(t)$	0.2	0	m ³ /s
$q_{m,out,BS}(t)$	0.2	0	m ³ /s
$\mu_{m,Ta,BS}(t)$	1	0	0-1 variable
$P_{i,REScur,BS}(t)$	$P_{i,RES,BS}(t)$	0	MW
$P_{i,G}(t)$ (IEEE-33)	10	0	MW
$P_{i,G}(t)$ (IEEE-123)	200	0	MW
$Q_{i,G}(t)$ (IEEE-33)	10	-10	MW
$Q_{i,G}(t)$ (IEEE-123)	200	-200	MW
$q_{m,S}(t)$ (8-node)	—	0	m ³ /s
$q_{m,S}(t)$ (24-node)	—	0	m ³ /s
$\omega_{mn,U}(t)$	1	0	p.u.
$P_{i,PVCur}(t)$	$P_{i,PV}(t)$	0	MW
$P_{i,WGCur}(t)$	$P_{i,WG}(t)$	0	MW

References

- Maupin MA, Kenny JF, Hutson SS, et al. Estimated use of water in the United States in 2010. U.S. Department of the Interior; 2014.
- The United Nations world water development report 2019-leaving no one behind. UNESCO World Water Assessment Programme (WWAP); 2019.
- Hamiche AM, Stambouli AB, Flazi S. A review of the water-energy nexus. *Renew Sustain Energy Rev* 2016;65:319–31. <https://doi.org/10.1016/j.rser.2016.07.020>.
- Vilanova MN, Balestieri JP. Exploring the water-energy nexus in Brazil: the electricity use for water supply. *Energy* 2015;85:415–32. <https://doi.org/10.1016/j.energy.2015.03.083>.
- Xiang X, Jia S. China's water-energy nexus: assessment of water-related energy use. *Resour Conserv Recycl* 2019;144:32–8. <https://doi.org/10.1016/j.resconrec.2019.01.009>.
- Kzltan M. Water-energy nexus of Turkey's municipalities: evidence from spatial panel data analysis. *Energy* 2021;226:120347. <https://doi.org/10.1016/j.energy.2021.120347>.
- Mancarella P. MES (multi-energy systems): an overview of concepts and evaluation models. *Energy* 2014;65:1–17. <https://doi.org/10.1016/j.energy.2013.10.041>.
- Agiwal M, Roy A, Saxena N. Next generation 5G wireless networks: a comprehensive survey. *IEEE Communications Surveys & Tutorials* 2016;18:1617–55. <https://doi.org/10.1109/COMST.2016.2532458>.
- Palattella MR, Dohler M, Grieco A, Rizzo G, Torsner J, Engel T, et al. Internet of Things in the 5G era: enablers, architecture, and business models. *IEEE J Sel Area Commun* 2016;34:510–27. <https://doi.org/10.1109/JSAC.2016.2525418>.
- Wang Q, Zhao X, Lv Z, Ma X, Zhang R, Lin Y. Optimizing the ultra-dense 5G base stations in urban outdoor areas: coupling GIS and heuristic optimization. *Sustain Cities Soc* 2020;63:102445. <https://doi.org/10.1016/j.scs.2020.102445>.
- Auer G, Giannini V, Desset C, Godor I, Skillermark P, Olsson M, et al. How much energy is needed to run a wireless network? *IEEE Wirel Commun* 2011;18:40–9. <https://doi.org/10.1109/MWC.2011.6056691>.
- Hui H, Ding Y, Shi Q, Li F, Song Y, Yan J. 5G network-based Internet of Things for demand response in smart grid: a survey on application potential. *Appl Energy* 2020;257:113972. <https://doi.org/10.1016/j.apenergy.2019.113972>.
- Israr A, Yang Q, Li W, Zomaya AY. Renewable energy powered sustainable 5G network infrastructure: opportunities, challenges and perspectives. *J Netw Comput Appl* 2021;175. <https://doi.org/10.1016/j.jnca.2020.102910>.
- The balance between 5G profit and power. Available at: <https://www.datacenterdynamics.com/en/analysis/balance-between-5g-profit-and-power/>.
- Dahal MS, Shrestha JN, Shakya SR. Energy saving technique and measurement in green wireless communication. *Energy* 2018;159:21–31. <https://doi.org/10.1016/j.energy.2018.06.066>.
- Li YR, Chen M, Xu J, Tian L, Huang K. Power saving techniques for 5G and beyond. *IEEE Access* 2020;8:108675–90. <https://doi.org/10.1109/ACCESS.2020.3001180>.
- H S. Green power for mobile Bi-annual report. GSMA; 2014.
- Hu S, Chen X, Ni W, Wang X, Hossain E. Modeling and analysis of energy harvesting and smart grid-powered wireless communication networks: a

- contemporary survey. *IEEE Trans Green Commun Netw* 2020;4:461–96. <https://doi.org/10.1109/TGCN.2020.2988270>.
- [19] Nortershauser D, Le Masson S, Volkov T, Galkin T, Amper O, Huttunen J. Experimental liquid cooled base station. In: *IEEE international telecommunications energy conference (INTELEC)*; 2016. p. 1–7. <https://doi.org/10.1109/INTELEC.2016.7749130>.
 - [20] Sun R. A novel concept methodology of in-direct cold plate liquid cooling design for data center in CPU socket area. In: *19th IEEE intersociety conference on thermal and thermomechanical phenomena in electronic systems (ITherm)*; 2020. <https://doi.org/10.1109/INTLEC.2016.7749130>.
 - [21] Nokia selected as nationwide supplier of 5G RAN by Elisa Finland. Available at: <https://www.nokia.com/about-us/news/releases/2020/09/30/nokia-selected-as-nationwide-supplier-of-5g-ran-by-elisa-finland/>.
 - [22] Khosravi A, Laukkanen T, Vuorinen V, et al. Waste heat recovery from a data center and 5G smart poles for low-temperature district heating network. *Energy* 2020;218:119468. <https://doi.org/10.1016/j.energy.2020.119468>.
 - [23] Rose I. Analysis of a water tank energy storage system for use in a warm water cooled data center. In: *15th IEEE intersociety conference on thermal and thermomechanical phenomena in electronic systems (ITherm)*; 2016. <https://doi.org/10.1109/ITHERM.2016.7517694>.
 - [24] Zamzam AS, Dall'Anese E, Zhao C, Taylor JA, Sidiropoulos ND. Optimal water—power flow-problem: formulation and distributed optimal solution. *IEEE Transactions on Control of Network Systems* 2019;6:37–47. <https://doi.org/10.1109/TCNS.2018.2792699>.
 - [25] Song SH, Mu YF, Meng XJ, Jia HJ, Xu J, Yuan K. Joint economical dispatch method of power distribution network and water distribution system for distributed generator accommodation. *Autom Electr Power Syst* 2019;43:95–115. <https://doi.org/10.7500/AEPS20180922001>.
 - [26] Zuloaga S, Khatavkar P, Vittal V, Mays LW. Interdependent electric and water infrastructure modeling, optimization and control. *IET Energy Systems Integration* 2019. <https://doi.org/10.1049/iet-esi.2019.0039>.
 - [27] Oikonomou K, Parvania M. Optimal coordination of water distribution energy flexibility with power systems operation. *IEEE Trans Smart Grid* 2019;10:1101–10. <https://doi.org/10.1109/TSG.2018.2824308>.
 - [28] Oikonomou K, Parvania M. Optimal coordinated operation of interdependent power and water distribution systems. *IEEE Trans Smart Grid* 2020;11:4784–94. <https://doi.org/10.1109/TSG.2020.3000173>.
 - [29] Sui Q, Wei F, Zhang R, Lin X, Tong N, Wang Z, et al. Optimal use of electric energy oriented water-electricity combined supply system for the building-integrated-photovoltaics community. *Appl Energy* 2019;247:549–58. <https://doi.org/10.1016/j.apenergy.2019.04.059>.
 - [30] Moazeni F, Khazaei J. Dynamic economic dispatch of islanded water-energy microgrids with smart building thermal energy management system. *Appl Energy* 2020;276:115422. <https://doi.org/10.1016/j.apenergy.2020.115422>.
 - [31] Roustaei M, Niknam T, Salari S, Chabok H, Sheikh M, Kavousi-Fard A, Aghaei J. A scenario-based approach for the design of smart energy and water hub. *Energy* 2020;195:116931. <https://doi.org/10.1016/j.energy.2020.116931>.
 - [32] Li C, Yao Y, Xie K, Hu B, Niu T. Integrated electrical, heating, and water distribution system to accommodate wind power. *IEEE Trans Sustain Energy* 2020;12:1. <https://doi.org/10.1109/TSTE.2020.3034134>.
 - [33] Zhao P, Gu C, Cao Z, Ai Q, Xiang Y, Ding T, et al. Water-energy nexus management for power systems. *IEEE Trans Power Syst* 2020;1. <https://doi.org/10.1109/TPWRS.2020.3038076> (early access).
 - [34] Huang X, Han T, Ansari N. Smart grid enabled mobile networks: jointly optimizing BS operation and power distribution. *IEEE Trans Netw* 2017;25:1832–45. <https://doi.org/10.1109/TNET.2017.2655462>.
 - [35] Labidi W, Chahed T, Elayoubi S. Optimal battery management strategies in mobile networks powered by a smart grid. *IEEE Trans Green Commun Netw* 2018;2:859–67. <https://doi.org/10.1109/TGCN.2018.2806299>.
 - [36] Liu X, Ansari N. Profit-driven user association and smart grid energy transfer in green cellular networks. *IEEE Trans Veh Technol* 2019;68:10111–20. <https://doi.org/10.1109/TVT.2019.2934416>.
 - [37] Renga D, Al Haj Hassan H, Meo M, Nuaymi L. Energy management and base station on/off switching in green mobile networks for offering ancillary services. *IEEE Trans Green Commun Netw* 2018;2:868–80. <https://doi.org/10.1109/TGCN.2018.2821097>.
 - [38] Al Haj Hassan H, Renga D, Meo M, Nuaymi L. A novel energy model for renewable energy-enabled cellular networks providing ancillary services to the smart grid. *IEEE Trans Green Commun Netw* 2019;3:381–96. <https://doi.org/10.1109/TGCN.2019.2893203>.
 - [39] Malik MU, Ali M, Tao J. The impact of Internet of Things supported by emerging 5G in power systems: a review. *Journal of Power and Energy Systems* 2019. <https://doi.org/10.17775/CSEEJPES.2019.01850>.
 - [40] Frenger P, Tano R. More capacity and less power: how 5G NR can reduce network energy consumption. In: *2019 IEEE 89th vehicular Technology conference, VTC2019-Spring*; 2019. <https://doi.org/10.1109/VTCSpring.2019.8746600>.
 - [41] Dahal MS, Shrestha JN, Shakya SR. Energy saving technique and measurement in green wireless communication. *Energy* 2018;159:21–31. <https://doi.org/10.1016/j.energy.2018.06.066>.
 - [42] Han D, Zheng B, Chen Z. Sleep mechanism of base station based on minimum energy cost. *Wireless Commun Mobile Comput* 2018;2018:1–13. <https://doi.org/10.1155/2018/4202748>.
 - [43] Wang LY, Chen QX, He GN, Zhong GB, Kang CQ. Optimization of generation scheduling considering battery energy storage life model. *Autom Electr Power Syst* 2019;43:93–101. <https://doi.org/10.7500/AEPS20180329002>.
 - [44] Baran ME, Wu FF. Network reconfiguration in distribution systems for loss reduction and load balancing. *IEEE Trans Power Deliv* 1989;2:1401–7. <https://doi.org/10.1109/61.25627>.
 - [45] Walski TM, Chase DV, Savic DA, Grayman W, Beck S, Koelle E. *Advanced water distribution modeling and management*. 2003.
 - [46] IEWS Data. Available at: <https://www.researchgate.net/project/Coordinated-operation-of-the-integrated-electricity-water-system-and-5G-base-stations/update/6064acb1220bc500014d2169>.
 - [47] Amaral Lopes R, Grønborg Junker R, Martins J, Murta-Pina J, Reynders G, Madsen H. Characterisation and use of energy flexibility in water pumping and storage systems. *Appl Energy* 2020;277:115587. <https://doi.org/10.1016/j.apenergy.2020.115587>.
 - [48] Bertsimas D, Tsitsikilis J. *Introduction to linear optimization*. Athena Scientific 1998. <https://doi.org/10.1038/258429a0>.
 - [49] Farivar M, Clarke CR, Low SH, Chandy KM. Inverter VAR control for distribution systems with renewables. In: *IEEE international conference on smart grid communications*; 2011. <https://doi.org/10.1109/SmartGridComm.2011.6102366>.
 - [50] Gan L, Li N, Topcu U, Low SH. Exact convex relaxation of optimal power flow in radial networks. *IEEE Trans Automat Contr* 2015;60:72–87. <https://doi.org/10.1109/TAC.2014.2332712>.
 - [51] Lucien B, Andreas V, Spyros C. Second-order cone relaxations of the optimal power flow for active distribution grids: comparison of methods. *Int J Elec Power* 2021;127:106625. <https://doi.org/10.1016/j.ijepes.2020.106625>.
 - [52] PJM data miner. Available at: <https://www.pjm.com/markets-and-operations/ops-analysis.aspx>.
 - [53] Marsan MA, Chiaraviglio L, Ciullo D, Meo M. Optimal energy savings in cellular access networks. In: *2009 IEEE international conference on communications workshops*; 2009. <https://doi.org/10.1109/ICCW.2009.5208045>.
 - [54] House LW. *Water supply related electricity demand in California*. Demand Response Research Center; 2006.